HPS Gaussian-Process Resonance Search
Combined Global Search and Observed Limits

Analysis note v4.9.16: probability audit and signal echoes

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Scope of the result. Full 2015 and 2016 data, plus native 2021 10%, are combined over 19–250 MeV. The upper limits are pointwise, asymptotic 90% CL_s limits. Figure 1 now separates the observed fit from background-reference probability diagnostics. Its local probabilities are nominal asymptotic values. The GP scan-wide tails remain conditional on specified background spectra; they do not establish a final particle significance or calibrate interval coverage. The extraction fits and upper limits are unchanged.

1 What the combined calculation adds

At each mass, the datasets that cover that mass share one possible particle coupling. Their information is combined in a single likelihood. Membership follows the established ranges; the analysis does not choose whichever dataset gives the strongest-looking result.

The upper curve limits the allowed particle contribution at each mass. Below it, the signed fit root and nominal local probability describe the same likelihood used in the extraction displays. A separate diagnostic shows what happens when that root is compared with a particular background reference and when the full scan is taken into account. Those probability questions have different assumptions; they cannot be interchanged. The look-elsewhere calculation does not rescale the upper limits. The largest positive combined fit is at 66 MeV: $r = 2.760$, with nominal local $p_0 = 0.00289$. This is a selected local reference value, not a global discovery probability. At 76 MeV the fit is much smaller, $r = 0.166$. Its former extreme conditional Gaussian tail comes from comparing that value with a reference background whose fitted root is -8.700. Rebinning the extraction displays changed neither result. The audit below explains the resulting discontinuities and the limits of the probability interpretation.

Mass [MeV]	Active data
19–38	2015 full
39–49	2015 full + 2016 full
50–90	2015 full + 2016 full + 2021 10%
91–180	2016 full + 2021 10%
181–250	2021 10%

Table 1: The complete search uses 1 MeV spacing and established dataset ranges.

2 Full combined search

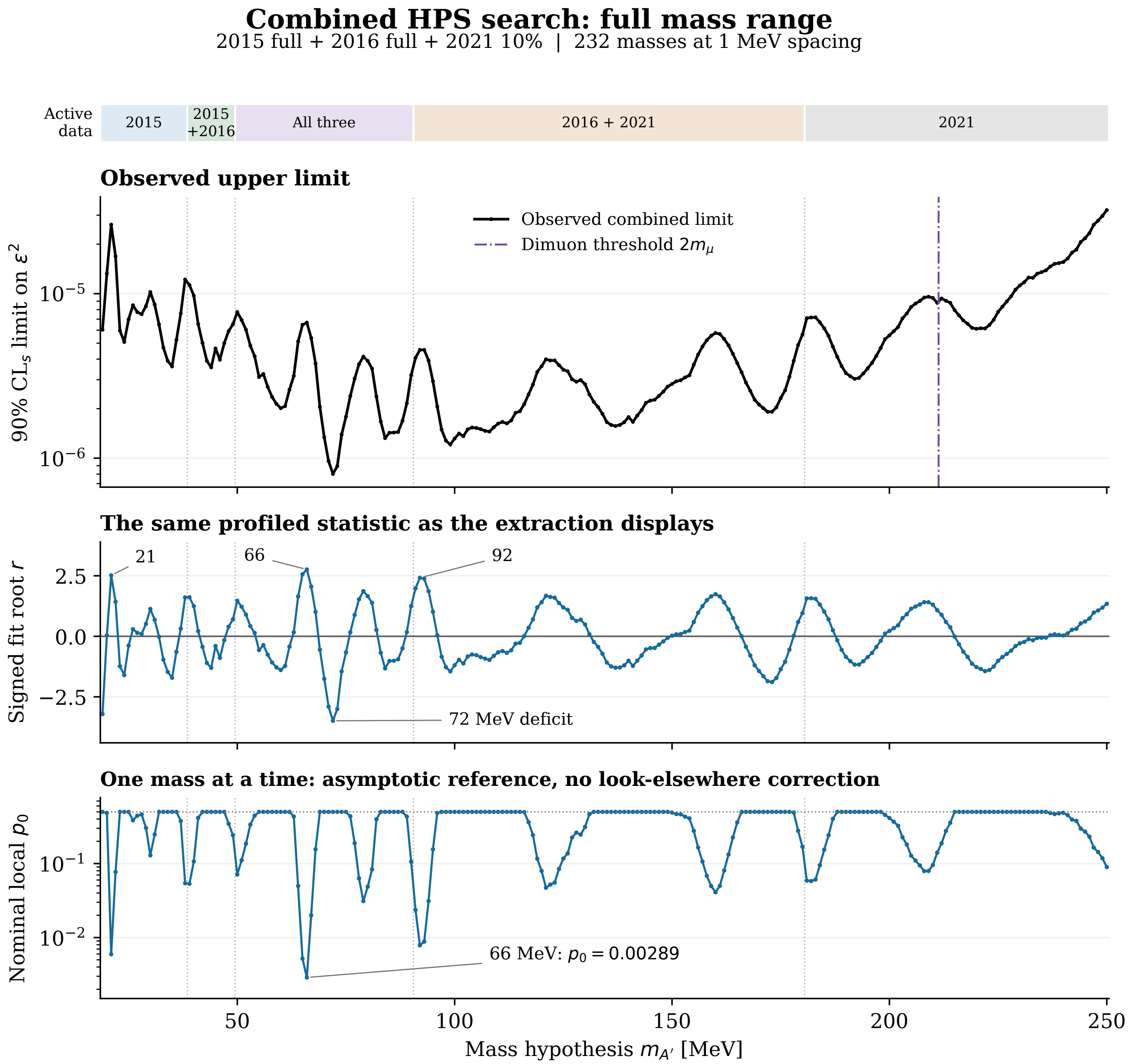

Figure 1: All 232 combined mass hypotheses are shown. Top: unchanged pointwise, asymptotic 90% CL_s limit, including the visible branching correction once above the dimuon threshold. Middle: observed signed profiled root r_m from the same native-bin likelihood as the extraction displays. Bottom: nominal local $p_0 = \overline{\Phi}[\max(0, r_m)]$, with the retained convention $p_0 = 0.5$ for nonpositive roots; this is neither a scan-wide probability nor a new calibration. The former "common-background Gaussian local" curve is separated into Figs. 2–3: it approximates the *sampling distribution of the root*, $r_m^* \sim N(a_m, s_m^2)$, under one fixed reference spectrum per year reused throughout the scan. "Common" means consistent across mass tests, and "Gaussian" describes the statistic's fluctuations, not a Gaussian-shaped mass background. That conditional curve is not another estimate of the local probability drawn here.

3 What the probability audit found

The fits are consistent; the reference distributions answer different questions. All 232 masses are present. Recomputing every saved probability and tail count reproduces the original scan. All 15 selected extraction roots agree with their source scans to below 10^{-15}. Display grouping only adds whole native bins; it cannot change a likelihood root. The apparently more significant extraction panels and the tiny conditional probabilities therefore did not arise from a new fit or a change of binning.

The earlier Figure 1 placed the nominal local curve beside a background-reference Gaussian local curve, then placed scan-wide tails underneath. Their labels did not sufficiently explain that they use different null references. The revised Figure 1 displays the unchanged upper limit, the observed root, and its nominal asymptotic local reference. Figures 2–3 retain the original conditional calculation as an explicit diagnostic. This presentation change does not substitute a newly calibrated significance or select a more favorable statistical test.

What "common-background Gaussian local" actually means

For each year, choose one complete generating background spectrum B_d. Use that same spectrum at every tested mass; different years retain their different spectra. The entire moving-mask training and signal fit is then applied to it. The fitted root at a mass can have a nonzero reference value $a_m = r_m(B)$, even though the generating spectrum has no added signal. Perturbing its bins gives a response width s_m and correlations across masses.

The Gaussian approximation concerns the resulting *statistic*, $r^*_m \simeq a_m + s_m Z_m$, not a Gaussian-shaped invariant-mass background. At one mass, Z_m has a standard-normal marginal. Across the scan, the Z_m are correlated. The retained local rule is

$$p_{\mathrm{ref}}(m) = \begin{cases} \overline{\Phi}[(r_m - a_m)/s_m]\,, & r_m > 0, \\ 1, & r_m \leq 0. \end{cases}$$

The principal global rule compares this observed threshold with the largest standardized score over positive-root masses in a complete background experiment. Under a correct reference distribution, this is a conditional one-sided test. Its probability is not the chance that a particle exists. Its numerical value need not equal the nominal asymptotic p_0 in Figure 1.

Why the plot jumped and why the 76 MeV tail looked extreme

There are 23 transitions between positive and nonpositive raw roots; 117 masses are assigned $p_{\mathrm{ref}} = 1$ by the sign rule. At 75 MeV, $r = -0.666$ and the reference is $a = -10.831$: the standardized score is 10.38, but the sign rule sets the probability to one. At 76 MeV, r becomes slightly positive, $+0.166$, while $a = -8.700$ and $s = 0.979$. The score is then 9.05, giving the formal Gaussian tail 6.98×10^{-20}. The corresponding nominal local value is 0.4342. The discontinuity is built into the declared sign rule plus the large moving reference offset; it is not evidence that the data suddenly acquire a large peak.

The rule is retained and its gated points are marked. Smoothing this jump or removing the sign requirement would change the test. The former global curve also broke where the GP sample had zero exceedances (76 and 77 MeV), and it showed direct-toy checks at only six representative masses. The revised diagnostics show every mass, with zero counts represented by upper bounds and sparse positive counts by open symbols and intervals.

Mass	r	a	s	Nominal p_0	Formal p_{mref}	GP global tails	Direct global
21	+2.516	+0.204	0.956	0.0059	0.0078	93598/200000	479/1000
66	+2.760	+8.987	0.980	0.0029	1.0000	200000/200000	1000/1000
72	-3.490	-7.160	0.978	0.5000	1.0000	200000/200000	1000/1000
75	-0.666	-10.831	0.979	0.5000	1.0000	200000/200000	1000/1000
76	+0.166	-8.700	0.979	0.4342	6.98×10^{-20}	0/200000	0/1000
78	+1.529	-1.839	0.980	0.0631	2.95×10^{-4}	6140/200000	26/1000
92	+2.416	-2.999	0.984	0.0078	1.89×10^{-8}	2/200000	0/1000

Table 2: An arithmetic audit, not a table of established particle significances. The formal column extrapolates a Gaussian marginal; global columns give actual exceedance counts for the conditional ordering. Nonpositive roots are gated to one. Counts and pointwise intervals for all 232 masses are in the probability ledger.

The defensible tail statement is limited by both the null model and validation. Zero of 1,000 independent direct spectra gives a one-sided 95% binomial upper bound of 0.00299 under that generating model. Zero of 200,000 GP fields gives 1.50×10^{-5} within the Gaussian approximation. Neither is a measured zero, and neither validates a 10^{-20} physical tail. At 92 MeV there are only two GP exceedances and no direct exceedance; the point estimate 10^{-5} has substantial Monte Carlo uncertainty. Agreement in the bulk or a satisfactory KS diagnostic cannot verify such far tails.

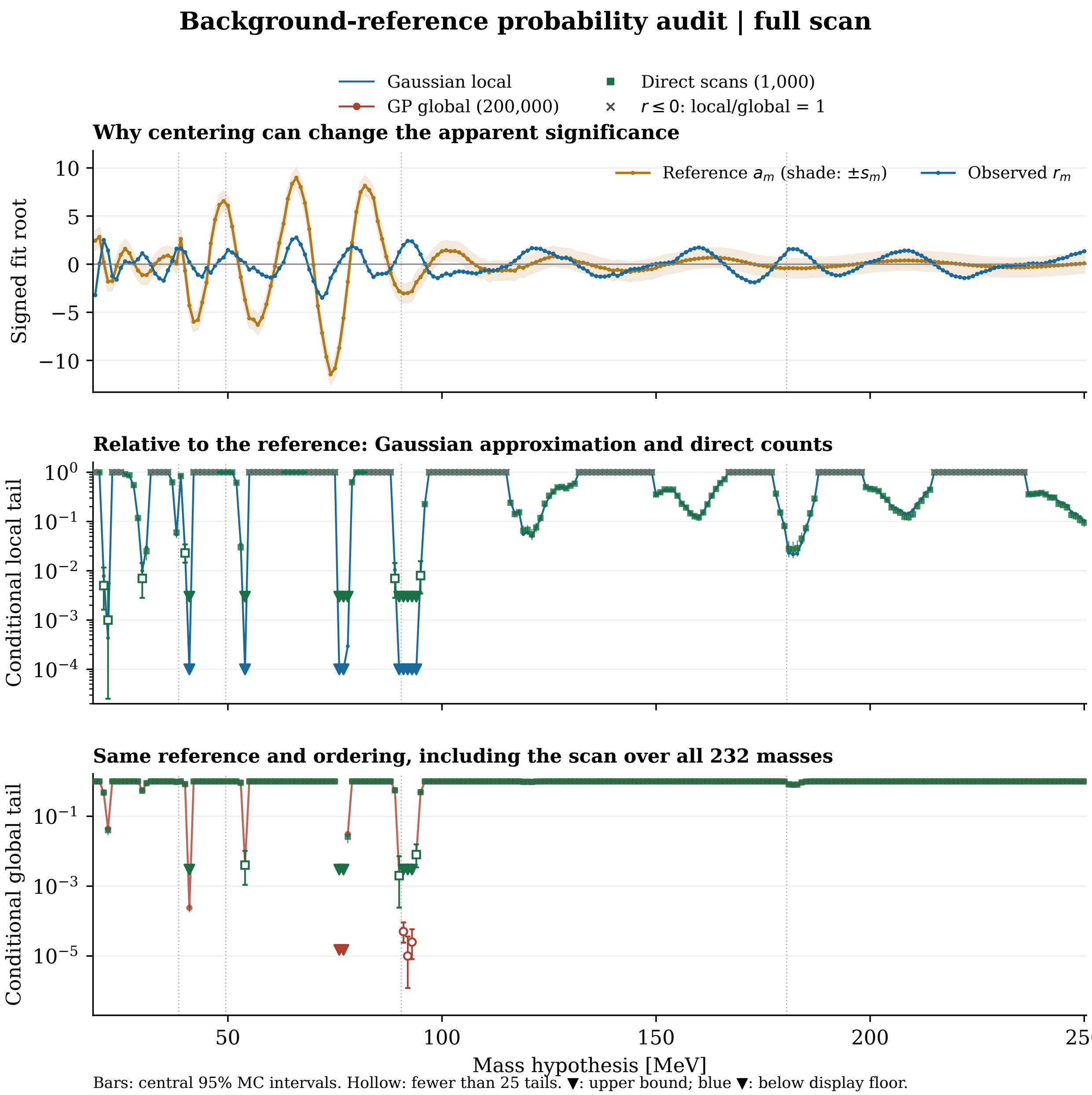

Figure 2: Complete conditional probability audit. Top: observed signed root and the background-reference offset; the shade is the response width, not an uncertainty band on the observed fit. Middle: the formal Gaussian local rule and direct local counts; crosses at one mark nonpositive roots. Bottom: global tails for the same reference-centered ordering using all 232 masses, with direct checks at every mass. Bars are pointwise central 95% binomial intervals. Open symbols have fewer than 25 exceedances. Red or green downward triangles denote one-sided 95% upper bounds after zero exceedances; blue triangles only indicate Gaussian local values below the 10^{-4} display floor. The two meanings are deliberately distinguished: no displayed floor is a measured probability. Correlation between mass points remains; intervals are not simultaneous.

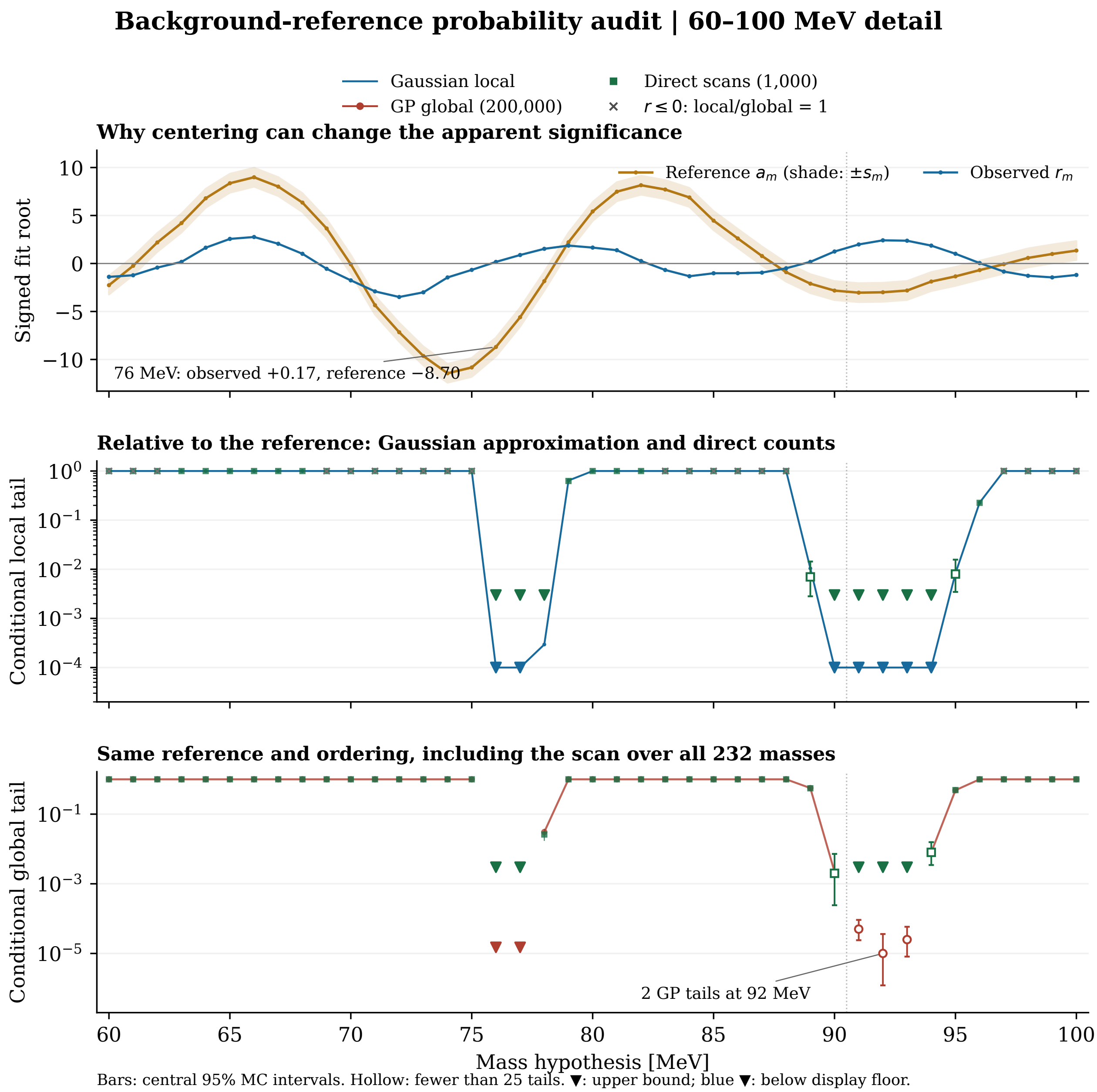

Figure 3: Detail of the same frozen scan and probability rules, without refitting or resampling. At 76 MeV the observed fitted signal is near zero, while the reference spectrum produces a large negative root. Centering against it creates the extreme formal local tail. At 75 MeV the observed root is negative, so the sign rule instead assigns one. The global search still covers the full 19–250 MeV grid; restricting the display to 60–100 MeV does not restrict the trials factor. At 92 MeV, two GP exceedances are shown with their interval rather than as a precise resolved tail. Symbols and interval conventions match Fig. 2.

Can the asymptotic profiled method support a final analysis?

Yes, profiling and asymptotic likelihood methods are defensible when their background model, constraints and sampling approximations are qualified [2]. This implementation has not established those conditions for a final discovery probability. The nominal curve assumes an appropriately centered, unit-width root under a suitable background null. The large offsets in the current reference construction show that this assumption does not hold for that construction. Subtracting those offsets answers a different conditional question; it does not, by itself, establish the appropriate physical null. Existing source-fit, 2016 numerical and development-sample qualifications still apply.

The GP trials-factor method [1] efficiently describes correlations of a suitably modeled significance field. Here it has been extended to retain nonzero offsets and measured response widths. It accelerates a declared probability calculation; it does not decide whether the chosen background is an adequate physical description. The very small conditional tails should be presented as tension with that reference, with explicit approximation and sampling limits, rather than as evidence for a particle. The separate raw-maximum test has tail estimate one for the observed maximum in both stored ensembles because the reference itself produces large positive roots elsewhere. This is another diagnostic of the reference and ordering, not a favorable global correction to choose in place of the first test.

The upper limits and apparent calibration changes should be judged separately. A weaker *observed* calibrated limit is not evidence for a loss of expected sensitivity. Expected sensitivity requires the relevant ensembles under qualified backgrounds, while coverage requires signal-plus-background hypotheses. Neither display rebinning nor the GP look-elsewhere calculation supplies that missing qualification.

4 Can a real peak make signal echoes?

Yes: the extraction procedure can produce them. Whether it explains the observed pattern remains unresolved. When the tested mass moves, its excluded signal region moves. A real signal hidden from the background training at its own mass can enter the sidebands for a neighboring test. The background GP then changes its prediction and correlated nuisance constraint. Relative to that changed background, the next fit can prefer a negative signal or a smaller positive signal elsewhere. Every added generating component can remain nonnegative throughout. A negative fitted residual is not a physical removal of events by the positive particle.

This is the *background-regression* GP. The second GP used for global significance emulates the correlated field of scan statistics. Accounting for correlations of background fluctuations in a trials factor does not automatically identify which observed peak, if any, generated other features under a signal alternative.

Replay with the current solver

The earlier 2021 peak–dip study used ten background and ten signal-plus-background toy spectra. Its paired 66 MeV injection shifted the median roots at 71 and 72 MeV by approximately -1.65 and -1.74. Its later reverse-injection study used a wider, smooth reference trained outside 60–86 MeV, with a kernel anchored at 66 MeV. The latter is a separate reference from the combined global-probability stress spectrum; the two studies must not be conflated.

We reused that wider smooth reference and the saved positive signal templates, then retrained and profiled with the *current dense solver*. There are 29 test masses (60–88 MeV), four deterministic generating spectra, and 29 reconstructions of the current observed spectrum. This adds 116 deterministic profile tests and no random toys. The full injected yields are 17,142 events at 66 MeV, 19,273 at 78 MeV, and 36,373 for the 65+78 MeV pair. These are selected illustrative strengths from the earlier single-signal fits, not an equal-strength comparison or a joint two-signal fit. The standalone low-mass injection is at 66 MeV, whereas the pair contains 65 MeV; the displayed curves are not an additivity test.

Subtracting the result for the same smooth background isolates the injection-induced change, $\Delta r(m) = r_m(B+S)-r_m(B)$. A positive 66 MeV injection gives negative changes at 71/72 MeV and positive changes near 78/80 MeV. A positive 78 MeV injection gives a smaller positive feature near 65/66 MeV and negative features near 71/72 and 85 MeV. Thus some features currently being inspected are precisely in regions where the analysis response can create echoes.

Test mass [MeV]	66 MeV injection	78 MeV injection	65+78 MeV injection
66	+2.292	+0.916	+2.839
71	-1.646	-1.825	-3.579
72	-1.740	-1.968	-3.385
78	+0.734	+2.707	+3.565
80	+0.829	+1.489	+2.180
85	-0.083	-1.833	-2.132

Table 3: Current-solver deterministic changes in the 2021 signed root, relative to the same background-only reconstruction. A change is a response diagnostic, not a significance, probability, or observed signal yield.

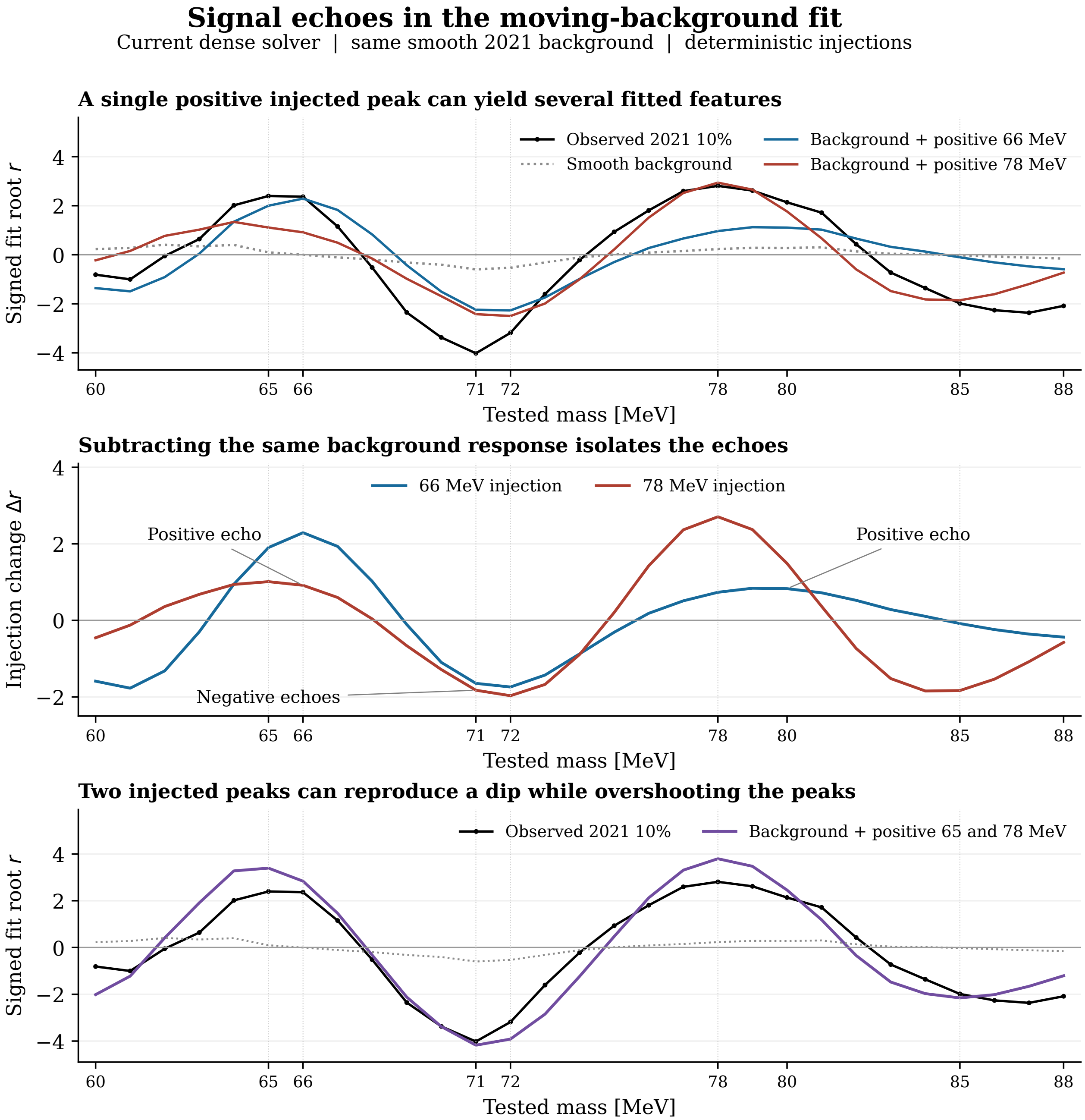

Figure 4: Signal echoes reproduced with the current dense 2021 solver and archived positive templates. Top: deterministic one-signal responses, the same background-only response, and the observed 2021 10% root. Middle: subtracting the background-only response isolates positive and negative echoes. Bottom: a 65+78 MeV injection can make a deep fitted deficit without a negative generating component, but the selected pair also produces peaks larger than observed. These are conditional mechanism demonstrations. No amplitudes were optimized to the current scan, no new data were opened, and the curves do not assign probabilities to one or two particles. The small differences from the old reconstruction (maximum $|\Delta r| = 0.0273$ for shared deterministic lanes) do not remove the echo pattern.

What the observed pattern does and does not establish

The selected full 65+78 MeV injection gives $r(71) = -4.177$, close to the observed -4.019. That agreement alone is incomplete: it predicts $r(65) = 3.395$ and $r(78) = 3.796$, above the observed 2.396 and 2.809, and a 72 MeV dip of -3.915 versus -3.186. The earlier three-quarter-strength pair reduced the peaks but also made the 71 MeV dip shallower. Those handpicked deterministic cases are not a likelihood comparison between one particle, two particles and background structure.

The stored combined background correlation study supports treating nearby extrema as dependent. For example, its GP correlations are $\rho(66, 72) = -0.689$ and $\rho(71, 78) = -0.661$; the corresponding direct-toy estimates are -0.693 and -0.638. The 66/78 MeV roots have positive correlation $+0.162$ in the GP approximation ($+0.119$ directly). These correlations concern fluctuations under the declared combined background. The injection replay concerns a change of generating mean in 2021 alone. They are complementary evidence for a correlated scan response, not interchangeable calculations and not proof of the same causal explanation in every year.

It is therefore plausible that some observed positive features or deficits are echoes of another feature. It is also plausible that a mismodeled continuum, detector/selection structure or correlated fluctuations produce them. The present results establish that an independent-particle interpretation of every extremum is unwarranted; they do not decide which physical explanation is correct. The extreme conditional tail at 76 MeV is separately explained by its reference offset and sign gate, not by an observed large 76 MeV signal.

How to test the echo explanation at 30%

Freeze a small family of parent-signal hypotheses and a background treatment before inspecting the next data increment. A useful comparison fits a single positive template near 65/66 MeV, a single template near 78 MeV, and a joint positive two-template model. Use a common signal-protected training interval that excludes both candidates for this comparison, qualify its interpolation with predeclared predictive controls, and retain each year's own response and normalization. Compare the complete native spectra and the scan pattern predicted after rerunning the moving-mask procedure. Separate one-template fits cannot supply the joint two-signal amplitude estimates or their covariance.

Generate or reuse properly matched paired background and signal-plus-background experiments for those declared hypotheses. Start with ten complete pilot spectra per hypothesis to check fit closure, response and runtime; those ten do not calibrate a far tail. Scale only after those checks pass, keeping full-spectrum mass correlations and nonoverlapping toy IDs. A formal preference must account for the mass and model choices already examined. Existing toys from a different generating spectrum cannot be relabeled as the new hypothesis.

Most decisively, test the additional disjoint 20% on its own before examining cumulative 30%. A parent peak and its echoes may all become visually stronger with more data; persistence of several extrema is not several independent confirmations. Predict their relative signs, shapes and fitted responses together. Background bias can also grow in units of statistical error. The staged-exposure displays below remain useful mean scenarios, but their simple signal overlays do not include a fresh background refit and therefore do not predict the full echo pattern. This revision does not unblind the 30% or 100% sample.

5 Illustrative scan of deficits

A negative auxiliary signal amplitude describes missing events with the signal-template shape; it is not a physical negative rate or coupling. This figure mirrors the excess scan without changing its fits or limits.

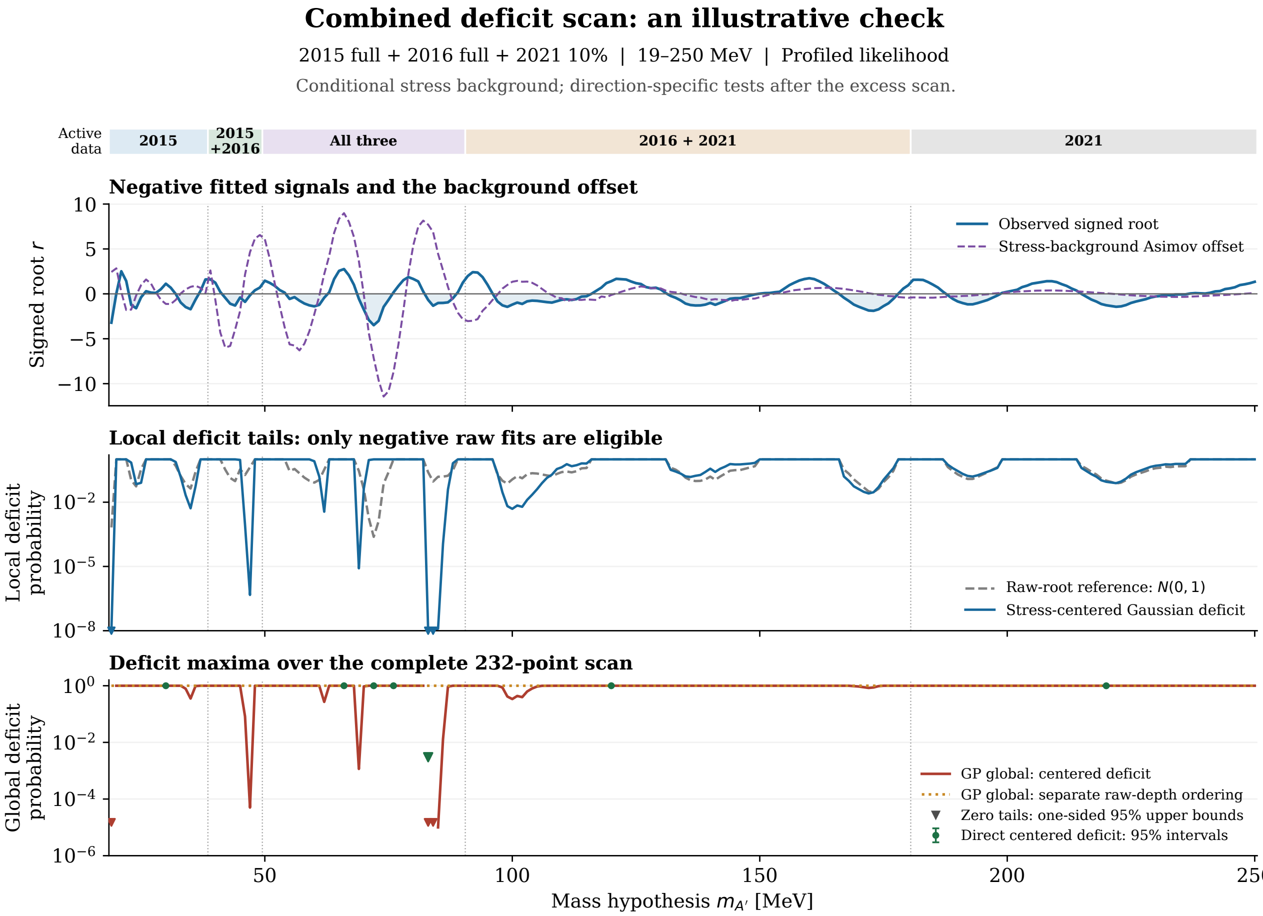

Figure 5: Profiled deficit scan. Top: observed signed roots and the stress-background offset; shading marks negative fitted contributions. Middle: raw-root Gaussian reference and conditional Gaussian-response tails. Both assign one to nonnegative raw roots. Local triangles mark the 10^{-8} display floor. Bottom: two separate global orderings; triangles mark one-sided 95% zero-count bounds. Direct error bars are central 95% intervals.

For $z = (r - a)/s$, the conditional local rule is $p^- = \Phi(z)$ when $r < 0$, and one otherwise. The principal scan score is $T^- = \max_{m:r_m<0}(-z_m)$, with $-\infty$ for an empty set. The separate raw-depth statistic is $D^- = \max_m \max(0, -r_m)$.

The deepest raw profiled deficit is at 72 MeV, with $r = -3.490$ and an uncalibrated local reference of 2.41×10^{-4}. The largest stress-centered deficit instead occurs at 83 MeV: $r = -0.676$, $a = +7.707$ and $s = 0.983$. Its raw reference is 0.2495, whereas the conditional local value is 7.18×10^{-18}. This contrast again comes mainly from the stress offset.

The latter threshold has 0/200,000 GP and 0/1,000 direct exceedances. The one-sided 95% global upper bounds are 1.50×10^{-5} and 0.003, respectively. Every simulated raw-depth maximum exceeds the observed deficit. These are the same realizations used for the excess scan; no new fits or independent toys were needed.

These illustrative tails follow the excess scan and do not adjust for choosing directions, methods or orderings. The unresolved tails establish neither a particle signal nor physical background validity.

6 Selected signal extractions and staged exposure checks

What the displays establish. The largest observed combined excess is at 66 MeV. All three years prefer a positive fitted signal there, although each estimate has substantial uncertainty. The next peak in the complete search is 21 MeV, where only 2015 contributes. The next leading peak covered by several datasets is 92 MeV, where 2016 and 2021 prefer noticeably different rates. These are useful locations for follow-up; the selected plots alone do not establish a particle or a failure of the background model.

The strongest raw combined deficit is at 72 MeV. Showing it alongside the excesses matters: the moving background fit can turn a positive injected signal into neighboring negative and positive fitted echoes (Sec. 4). A fitted deficit therefore does not by itself argue against a positive signal elsewhere. Background mismatch and correlated statistical fluctuations can also produce the pattern. The full response and an independent data increment are needed to distinguish these possibilities.

Which locations are shown

Selection uses the dense *observed profiled* signed root, whose ordering matches its one-sided asymptotic local reference on the positive branch. Select positive local maxima, ordered by decreasing root, with a separation greater than 2.25 times the larger resolution at the two coordinates. For the combined rule use the largest active resolution. Endpoints remain eligible and are identified. The stress-centered ordering is discussed separately; it does not choose the leading extraction panels.

Data	First excess	Second excess	Deepest deficit
2015 full	51 (+3.14)	21 (+2.52)	19 (-3.21)
2016 full	90 (+3.42)	117 (+3.28)	102 (-4.50)
2021 10%	78 (+2.81)	65 (+2.40)	71 (-4.02)
Full combined union	66 (+2.76)	21 (+2.52)	72 (-3.49)

Table 4: Selected mass in MeV, followed by the signed root in parentheses. These are selected local fit summaries, not global significances. The 19 MeV deficit is the 2015 search endpoint. The additional combined 92 MeV display has $r = +2.42$ and uses 2016 plus 2021.

How to read the figures

The top row shows event counts, the GP mean before profiling, the background fitted together with the signal, and their sum. The lower row subtracts that fitted background and overlays the fitted signal. Black bars show counting uncertainty only. The gray envelope is a zero-centered width guide from the original correlated GP constraint, projected into the display bins. It does not depict that constraint's center, an error band on the fitted background, or the full residual covariance. These residuals are not independent significance measurements.

Display bins are whole original bins grouped to approximately half the mass resolution, with a fixed histogram-edge origin. The likelihood retains every native fit bin. Multi-year sums use a common 1.25 MeV grid restricted to the overlap of the fitted windows. Each year keeps its own resolution, acceptance and count-to-coupling conversion in the likelihood. No fitted curve is extended outside its fit window.

The leading combined excess: 66 MeV

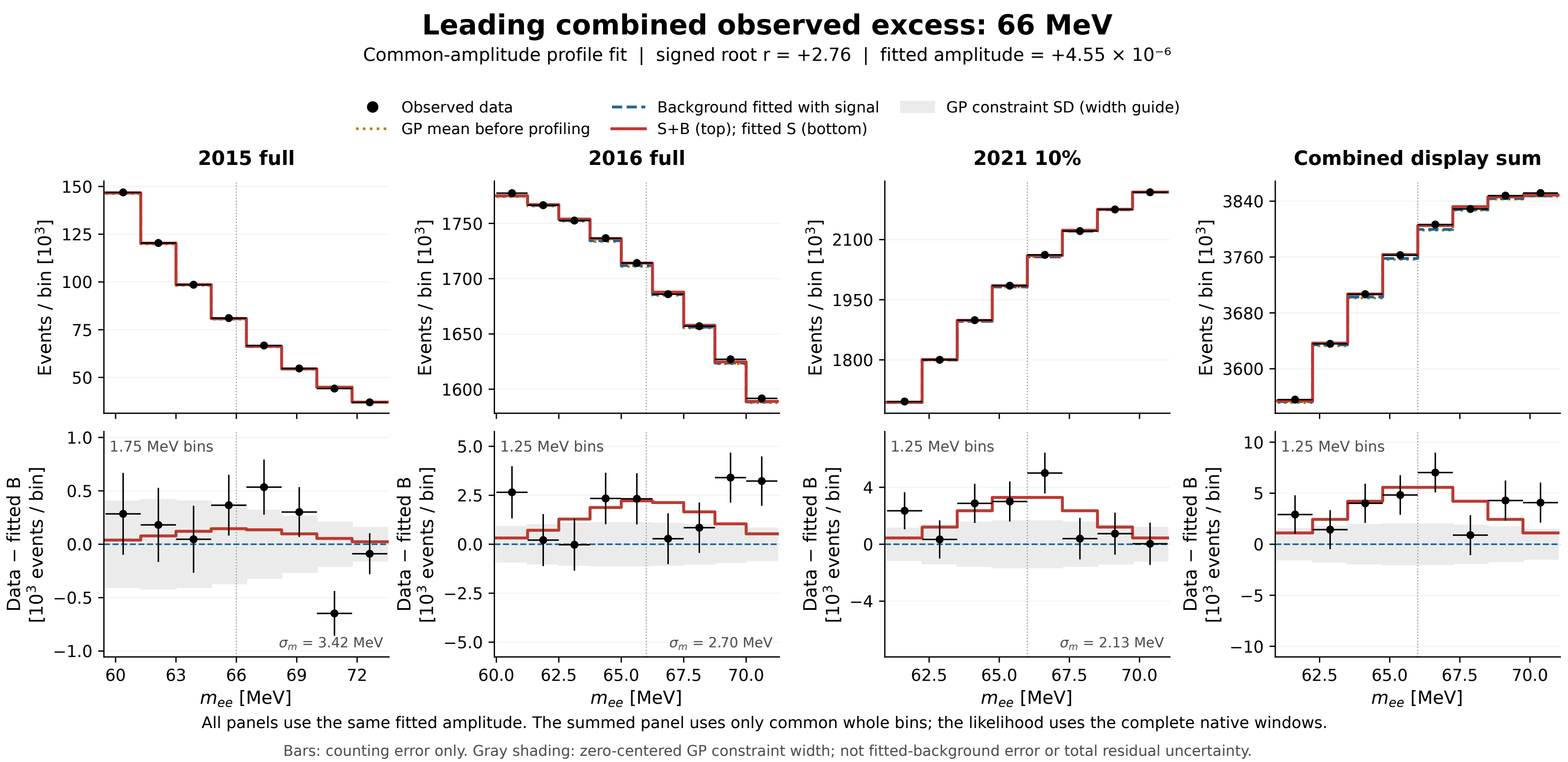

Figure 6: The 66 MeV common-amplitude fit, shown separately for each year and as a display sum. The shared signed amplitude is $\widehat{\epsilon}^2 = 4.55 \times 10^{-6}$ and $r = 2.76$. Every panel uses the same amplitude, with the appropriate year-specific yield and resolution. Summed panels omit boundary pieces; the likelihood root and quoted fitted yields use the full native windows.

A second location with multiple datasets: 92 MeV

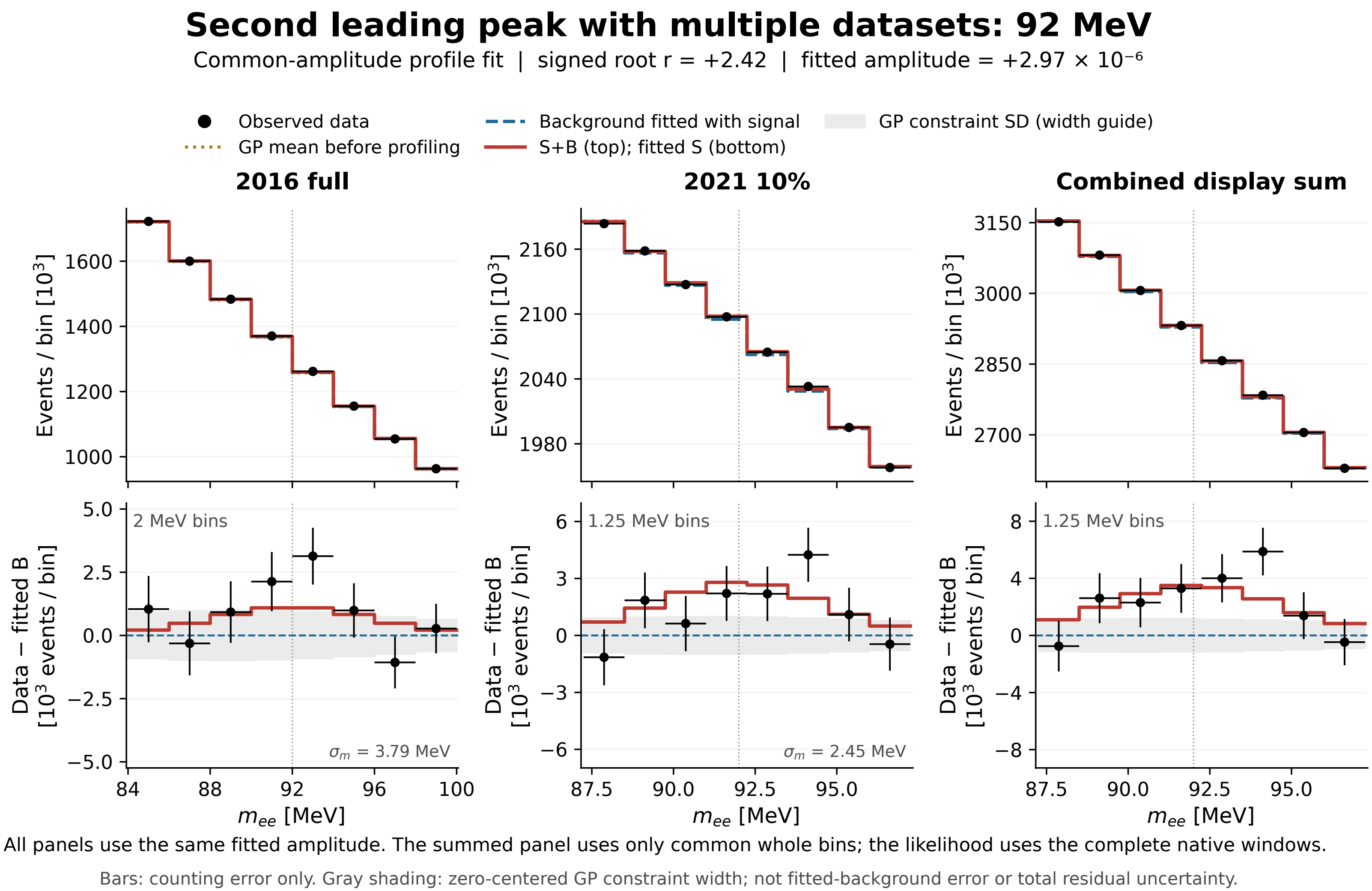

Figure 7: The 92 MeV fit uses 2016 and 2021. The common amplitude is $\hat{\epsilon}^2 = 2.97 \times 10^{-6}$ and $r = 2.42$. Both separate fits are positive, but their preferred rates differ (Fig. 13). A persistence illustration at this common rate is a testable assumption.

The deepest combined observed deficit: 72 MeV

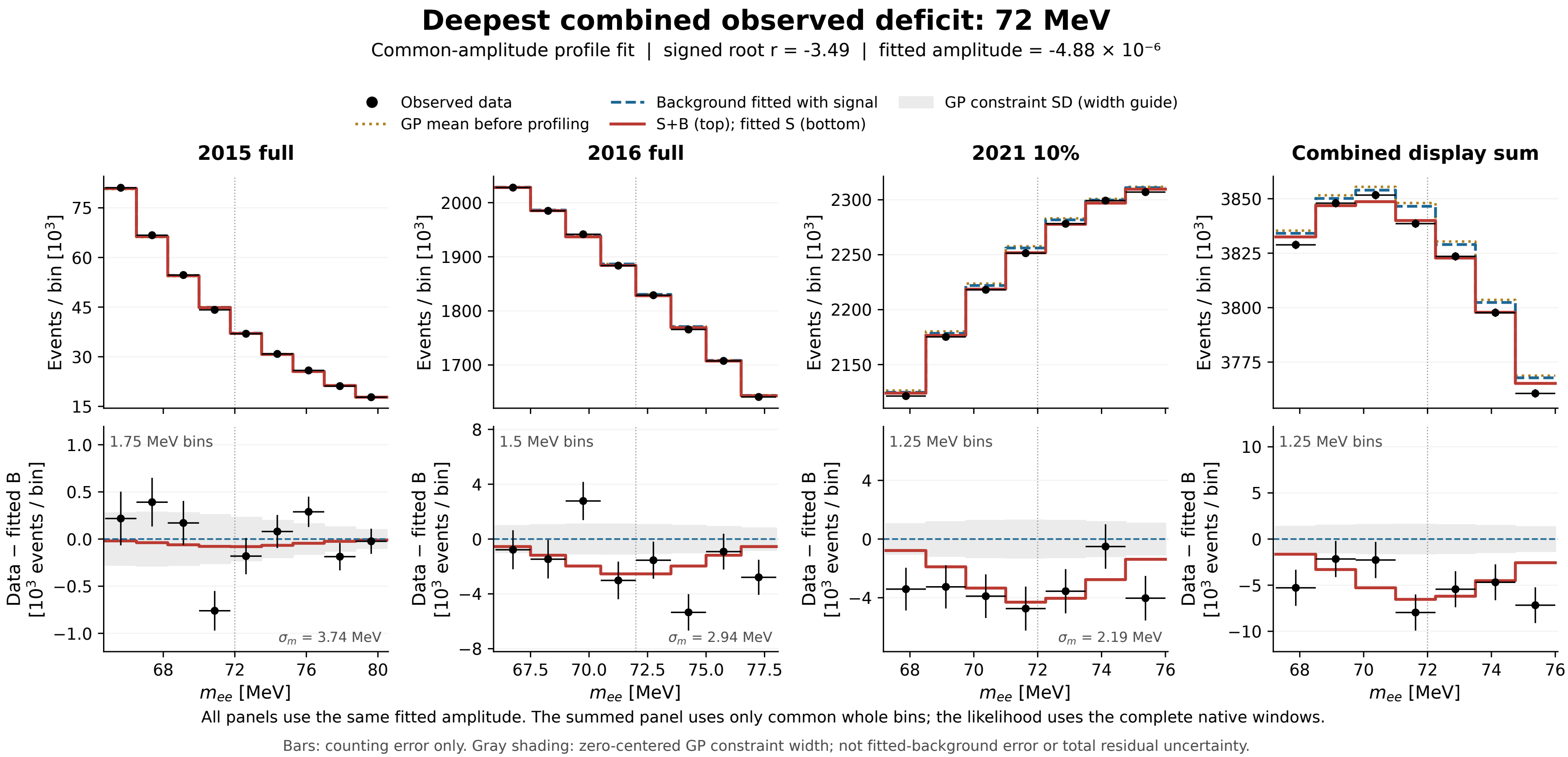

Figure 8: The deepest raw combined deficit has $r = -3.49$ and a signed auxiliary amplitude $\widehat{\epsilon}^2 = -4.88 \times 10^{-6}$. All three separate amplitudes are negative, with 2021 providing the strongest individual contribution. Negative templates describe missing events relative to the fitted background, not a physical negative coupling.

Individual signal extractions: 2015 full

The 51 MeV point is the largest observed profiled excess in 2015. The 21 MeV point is also the second leading peak of the full combined union, because 2015 is the only active dataset at this mass. It is only 2 MeV above the search endpoint; the 19 MeV deficit in Fig. 12 makes boundary and background-shape controls particularly relevant. More 2021 data cannot test this 21 MeV location.

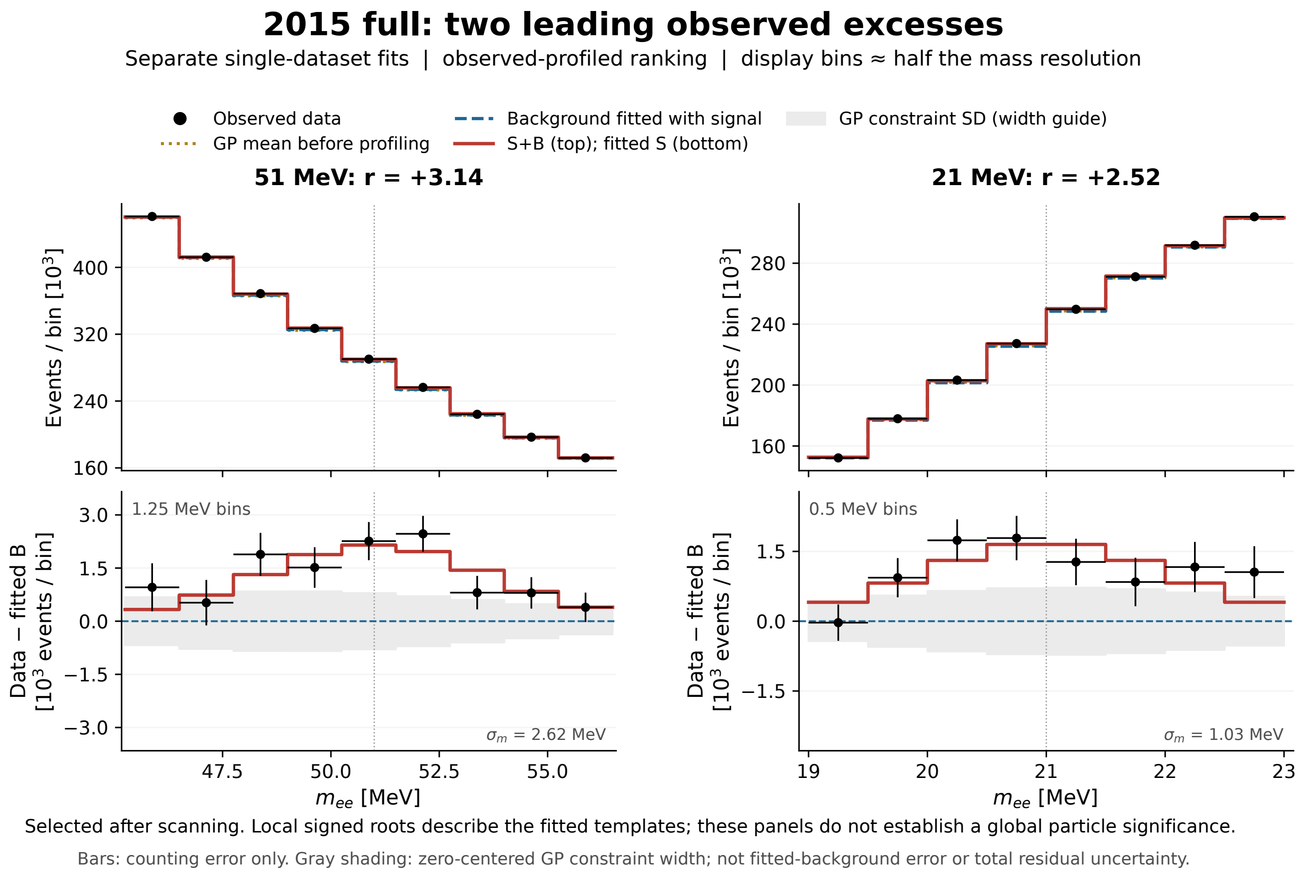

Figure 9: The two leading separated observed profiled excesses in 2015 full. Each panel is an independent single-dataset fit. Binning is chosen from the resolution and original histogram origin, not from the observed residual shape.

The lower panels are background-subtracted displays conditioned on the fitted template. A visible peak here should be assessed together with the background constraint, the fitted amplitude uncertainty, the scan selection and independent-data checks.

Individual signal extractions: 2016 full

The two leading individual 2016 peaks are at 90 and 117 MeV. Their positions differ from the leading combined mass because a common coupling also has to describe the other active dataset contributions. These panels retain the inherited 2016 source-fit waiver, development overlap and numerical qualifications; cleaner presentation does not resolve them.

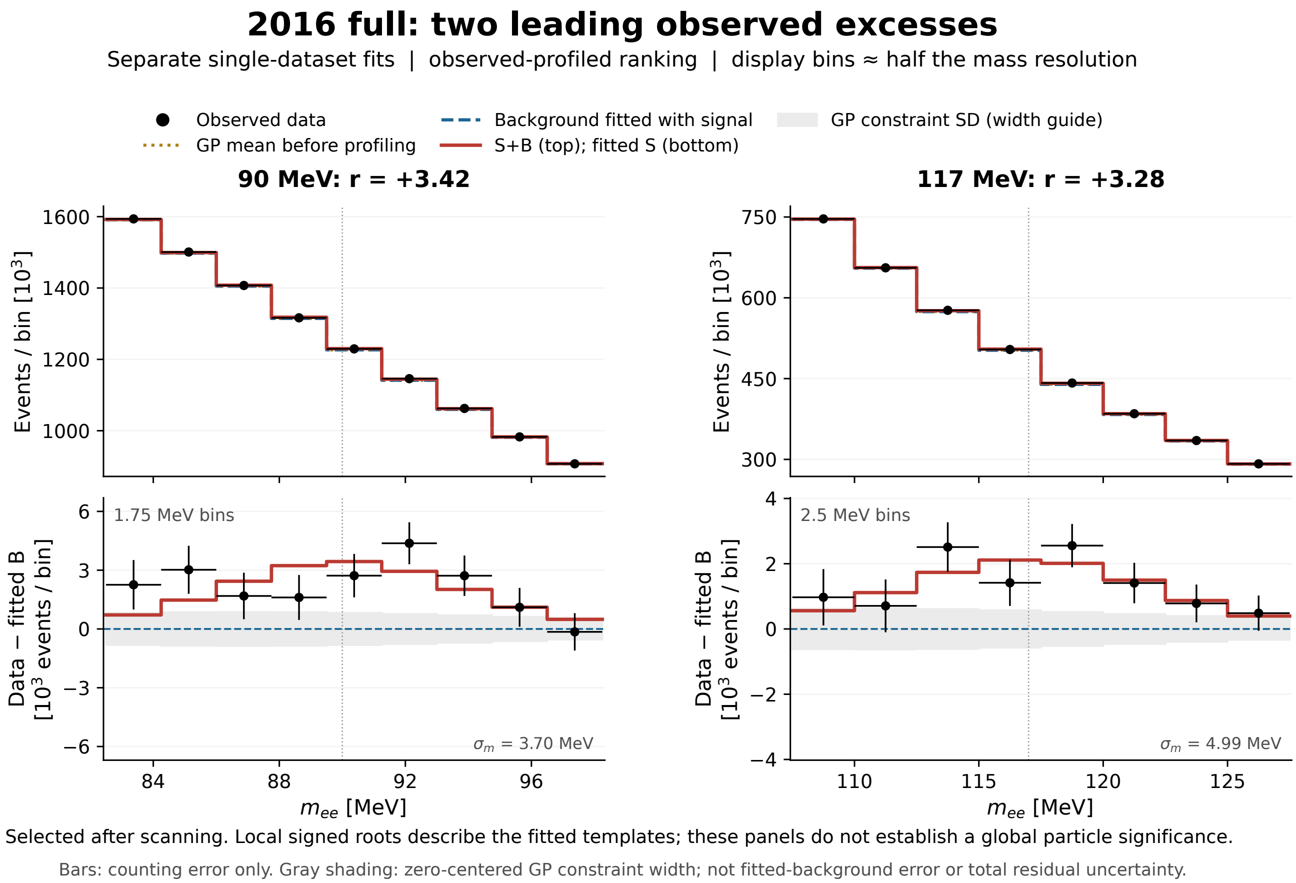

Figure 10: The two leading separated observed profiled excesses in 2016 full. Each panel is an independent single-dataset fit. Binning is chosen from the resolution and original histogram origin, not from the observed residual shape.

The lower panels are background-subtracted displays conditioned on the fitted template. A visible peak here should be assessed together with the background constraint, the fitted amplitude uncertainty, the scan selection and independent-data checks.

Individual signal extractions: 2021 10%

The released 2021 sample has leading individual peaks at 78 and 65 MeV. These are near, but not interchangeable with, the combined coordinates. Changing the mass or window after seeing a new sample would create another selection. Freeze these individual locations as a declared secondary family if they are followed at the 30% checkpoint.

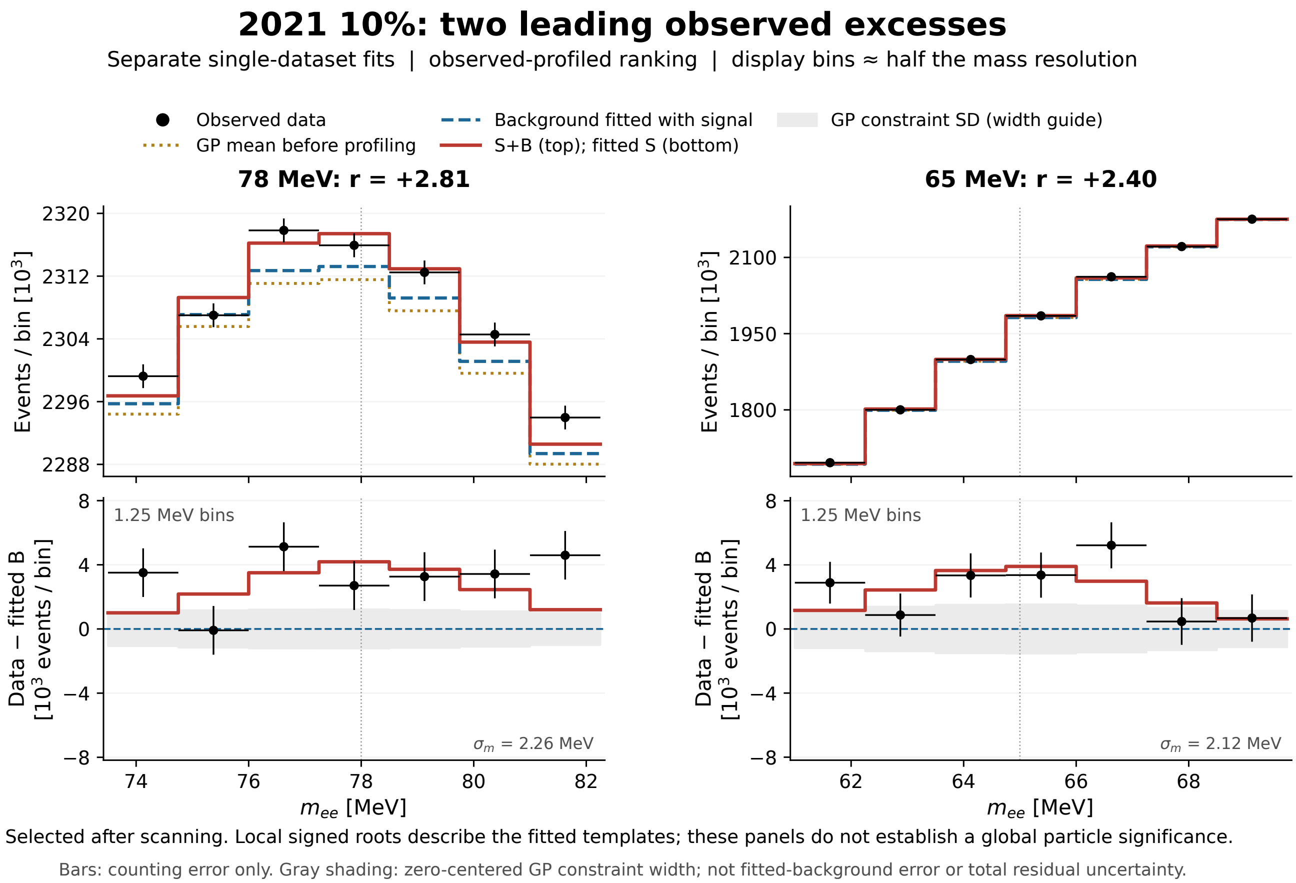

Figure 11: The two leading separated observed profiled excesses in 2021 10%. Each panel is an independent single-dataset fit. Binning is chosen from the resolution and original histogram origin, not from the observed residual shape.

The lower panels are background-subtracted displays conditioned on the fitted template. A visible peak here should be assessed together with the background constraint, the fitted amplitude uncertainty, the scan selection and independent-data checks.

Individual deficits as a background diagnostic

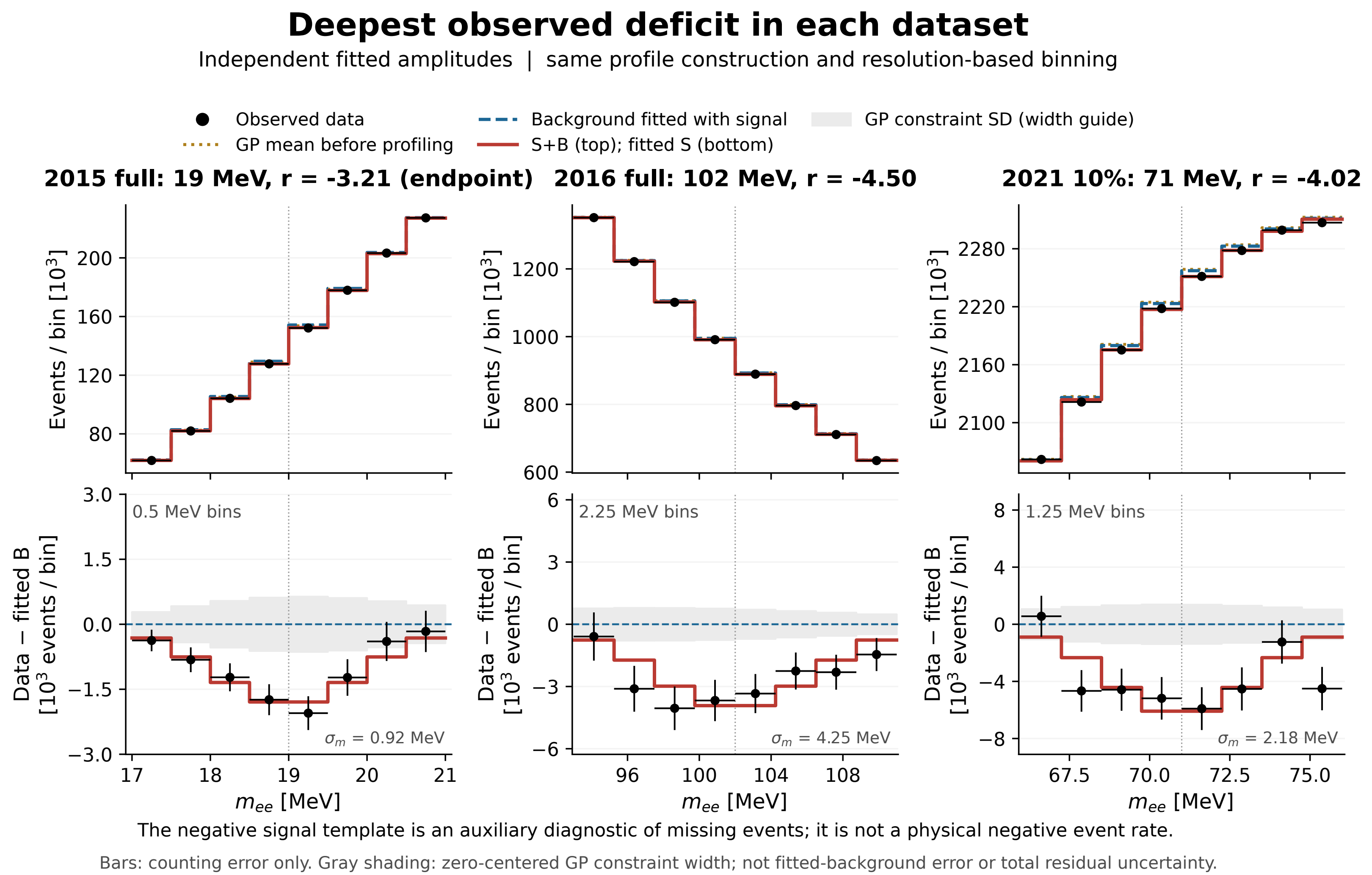

Figure 12: Deepest observed negative roots in 2015, 2016 and 2021. The 2015 point lies at the scan endpoint. The 2016 and 2021 minima are at 102 and 71 MeV, respectively. The negative branch provides a diagnostic with the same narrow template; no physical negative signal rate or calibrated deficit discovery is inferred.

Do separate years support the same rate?

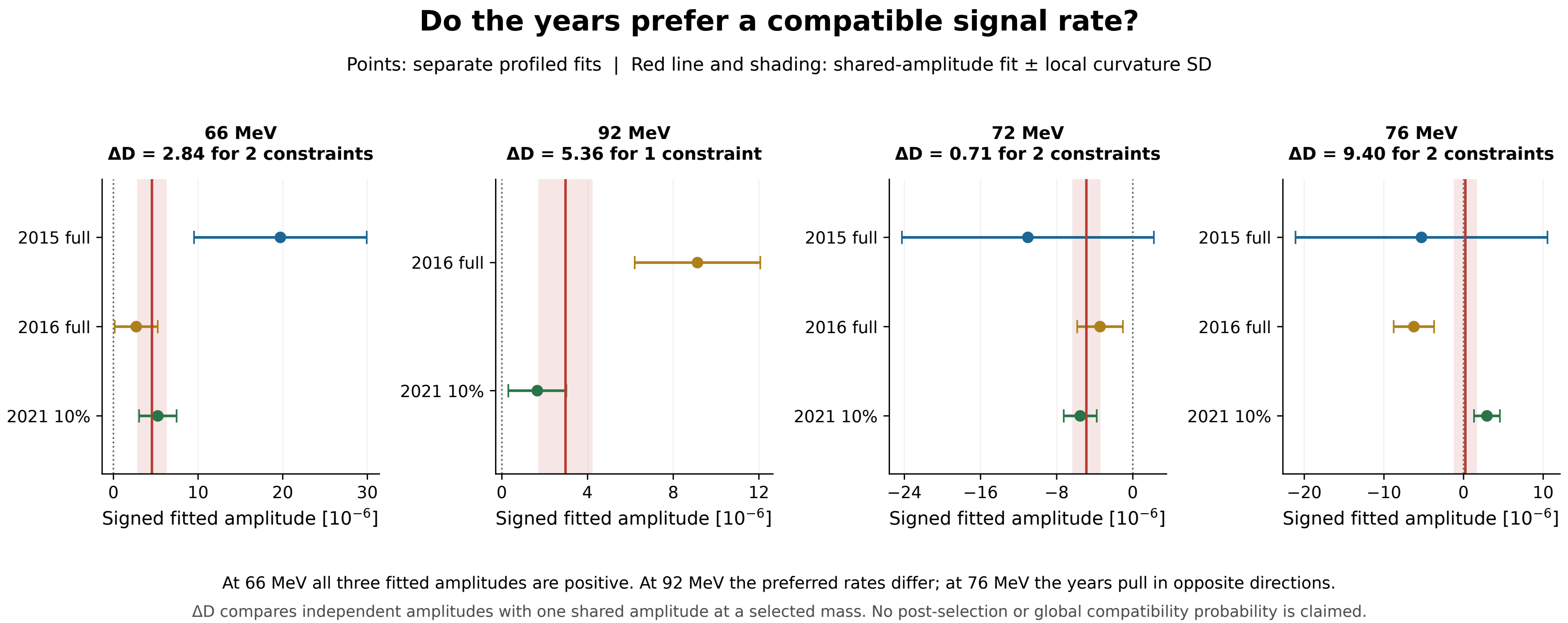

Figure 13: Independent signed amplitude estimates and local curvature standard errors, compared with the common fit. Define $\Delta D = 2[\mathrm{NLL}_{\mathrm{common}} - \sum_d \mathrm{NLL}_{d,\mathrm{free}}]$. The number of additional amplitude parameters is one fewer than the number of datasets. These masses were selected from the data, so the displayed likelihood losses are conditional descriptive comparisons; no post-selection compatibility probabilities are quoted.

Why the most extreme stress-centered tail need not look like a peak

The stress-background centering changes the reference distribution of the statistic; it does not change the observed fitted signal. At 76 MeV the combined raw root is only $+0.166$, but the generating stress construction has Asimov root $a = -8.700$ and response width $s = 0.979$. Subtracting this large negative offset produces a standardized value of 9.05. At 83 MeV the observed raw root is -0.676, with $a = +7.707$ and $s = 0.983$, giving a standardized deficit depth of 8.53.

The very small conditional tails therefore ask about departure from those particular stress constructions. They do not mean that the data contain a correspondingly large positive or negative fitted signal. At 76 MeV, 2016 prefers a negative amplitude while 2021 prefers a positive one. The common-versus-independent likelihood loss is $\Delta D = 9.40$ for two additional amplitudes. This argues against reading the conditional tail as evidence for a coherent resonance *at 76 MeV*; it does not exclude a background-fit echo of a signal at another mass. The reference offset and the echo mechanism are distinct effects, and neither has been identified as the physical origin of these data.

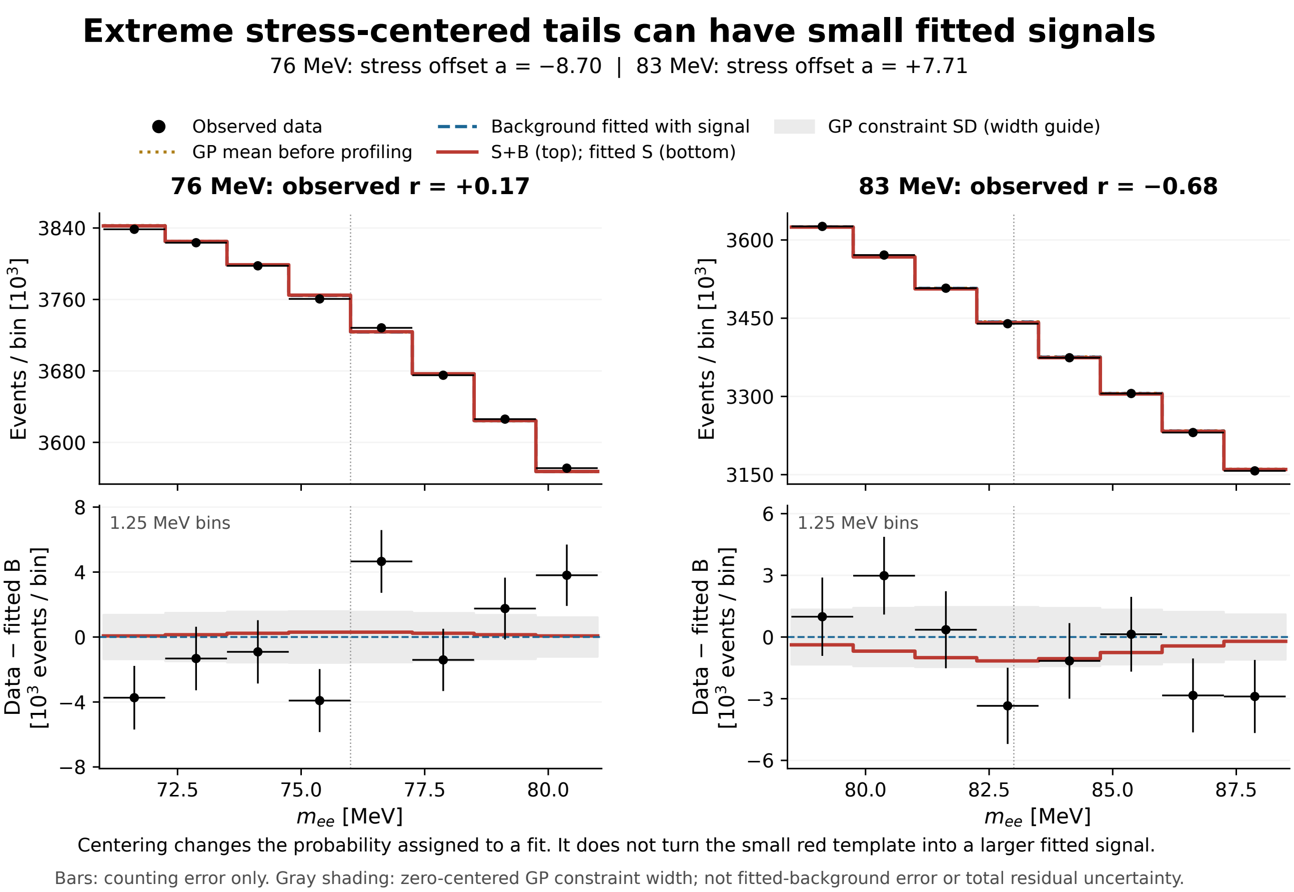

Figure 14: Common-window sums at the two principal stress-centered extrema. The small fitted signal templates are consistent with their small raw roots. Large stress offsets cause the extreme centered-tail interpretation. These sums have no independent fit.

The asymptotic profiled method is not invalid merely because calibration changes a probability or an observed limit. Its use in a final analysis requires qualification of the background family, the prediction uncertainty, signal response and the relevant sampling distributions. The present displays do not measure expected sensitivity. A weaker calibrated observed limit can reflect a correction to background behavior rather than information being removed from the data.

A 30% checkpoint before the full 2021 sample

Use the released 10% to define the hypotheses. Treat an additional, disjoint 20% as the next independent test of those fixed locations and rates. Only then inspect the cumulative 30% combination. This is more informative than a cumulative display alone, because the latter still contains the fluctuation that selected the peak. The same original-event membership and processing must be documented for every increment.

Let N_{10} be the observed first sample, B_{10} its background fitted in the selected common-amplitude model, and S_{10} the signal template at that fitted rate. Figure 15 compares two illustrative conditional means for a total exposure f times the original 10%:

$$\begin{aligned} \text{background only in added data:} \quad & N_{10} + (f-1)B_{10}, \\ \text{selected rate persists:} \quad & N_{10} + (f-1)(B_{10} + S_{10}). \end{aligned}$$

The independent additional 20% has means $2B_{10}$ or $2(B_{10} + S_{10})$. For the conditional cumulative 30% and 100% views the background-only counting variances from the future sample are $2B_{10}$ and $9B_{10}$, respectively. The observed first 10% is held fixed. These figures are mean scenarios, not new observations, toy experiments or discovery projections. They omit uncertainty in the assumed background and selected signal rate. The 92 MeV common-rate assumption is less consistent with the individual estimates than the 66 MeV assumption.

Mass [MeV]	10%	new 20%	30% total	100% total
66	14.6	29.2	43.7	145.8
92	13.6	27.2	40.9	136.2

Table 5: Illustrative mean signal yields in thousands of events in the complete 2021 fitted window if the selected common-fit rate persists. These are template yields, not the raw total counts or measured future yields. Display sums may omit boundary pieces.

A statistics-dominated precision reference can be computed without toys. With the original per-year signal yield vector u_d, GP constraint covariance C_d and GP mean b_d, define

$$I_d = u_d^{\top}[\mathrm{diag}(b_d) + C_d]^{-1}u_d, \qquad I(f) = I_{2015} + I_{2016} + fI_{2021}.$$

An absent dataset has $I_d = 0$. This local background-Asimov reference follows the usual information-scaling argument [2] and assumes that the *entire* 2021 count covariance scales as f, with unchanged acceptance and resolution. Fractional systematic effects or background bias need not improve this way. The factor $\sqrt{I(f)/I(1)}$ below is a precision illustration, not a multiplier for the observed combined root or a forecast global significance.

Mass [MeV]	Current 2021 information share	30% gain	100% gain
66	55.3%	1.45	2.44
92	82.4%	1.63	2.90

Table 6: Combined precision gain with 2015 and 2016 held fixed, under the stated covariance-scaling assumption. For 2021 alone the corresponding factors would be $\sqrt{3}$ and $\sqrt{10}$. There is no 2021 gain at 21 MeV.

Conditional exposure displays: 10%, new 20%, 30%, 100%

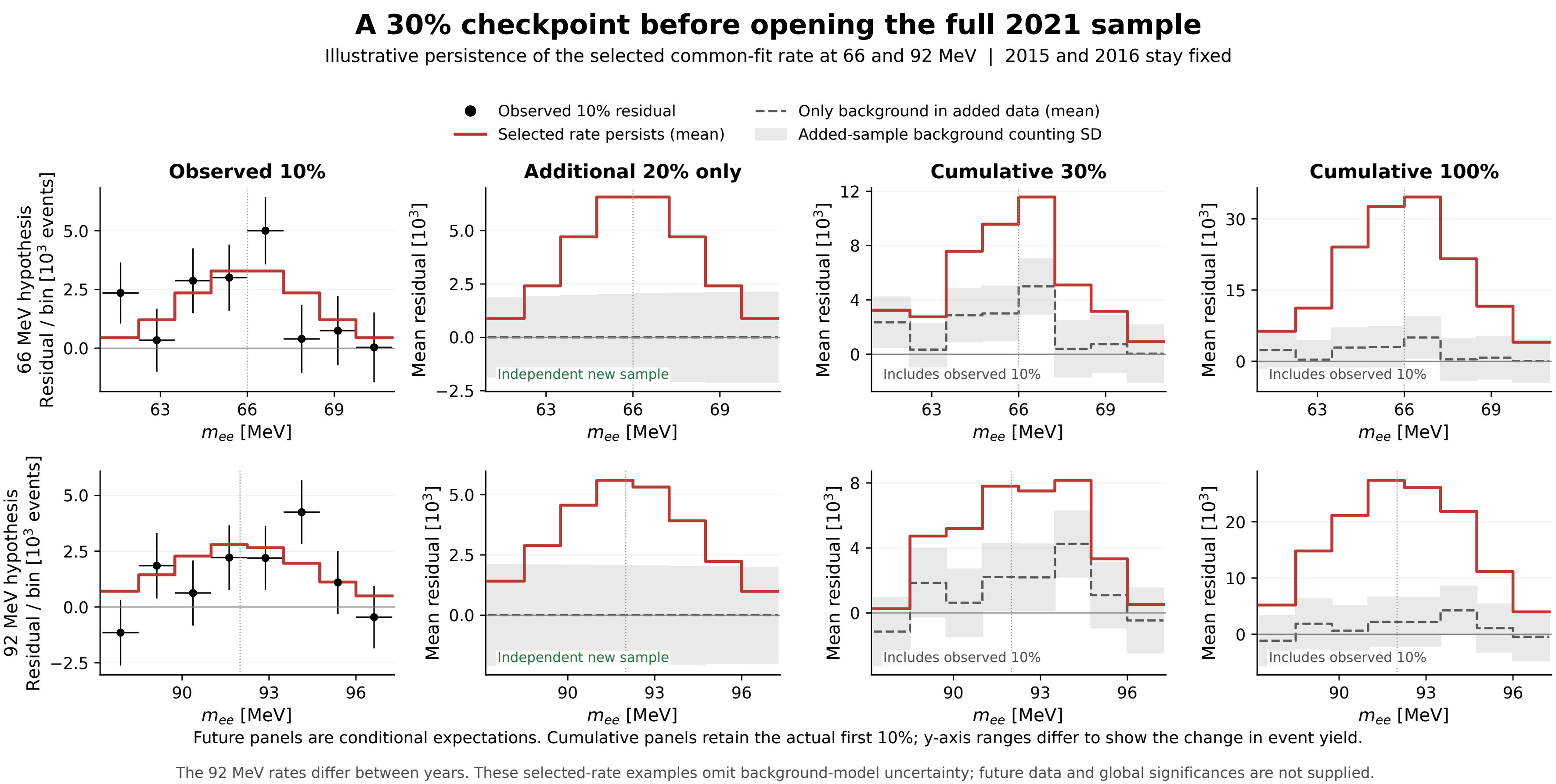

Figure 15: Background-subtracted 2021 views at fixed 66 and 92 MeV hypotheses. Only the first column contains observed points. Future curves keep the first sample fixed where included and add the mean of either hypothesis. Gray envelopes show only the background counting standard deviation of the added sample. They exclude fitted-background uncertainty and do not imply independent residual bins after a future refit. The vertical scales differ.

What would make the next comparison decisive?

First, define a disjoint additional 20% with the same selection, calibrations, mass resolution and normalization convention. Freeze the 66 and 92 MeV primary locations, and state in advance whether the individual 2021 locations at 65 and 78 MeV and the 71/72 MeV deficit checks are a secondary family. Do not reselect the mass from the new spectrum when reporting the fixed-location validation.

Fit the additional 20% on its own before combining it with the old 10% or the 2015/2016 samples. Compare its fitted rate, width behavior and background residuals with the stored predictions. A stable signal should reproduce a compatible rate, not merely increase the height of a cumulative display. Rate agreement alone would still leave detector effects or common background structure to investigate. A real parent signal and its fitted echoes may both grow with exposure, so growth of several extrema is not independent confirmation of several particles. Compare the complete predicted response, including the signs and relative amplitudes of echoes, in the additional 20% alone (Sec. 4).

Use predeclared sideband and predictive checks to test the background interpolation and relevant detector/selection effects, including both positive and negative residuals. The 2016 background-source transition and existing numerical/source qualifications remain part of that work. Model choices must not be selected by whichever one makes the observed peak more impressive.

The 30% and 100% cumulative looks share events. Before treating either as a formal discovery or exclusion result, declare the family of mass, direction and exposure tests and the treatment of sequential looks. Rebuild the appropriate signal and background sampling distributions for the new exposure and background family. The existing GP global-tail bank describes the current released sample and cannot calibrate a different exposure.

For an efficient follow-on calculation, begin with ten complete pilot spectra per declared generating hypothesis, retaining each same full spectrum throughout its mass scan. Use new deterministic seed namespaces and nonoverlapping toy IDs. Time the pilot, test exact-fit and approximation closure, then scale in restartable batches only if the validated cost is acceptable. Background-only scans address global discovery tails; calibrated limits additionally require signal-plus-background hypotheses. No new toys were necessary for the present extraction and mean-scenario displays.

Display and reconstruction checks

Fifteen selected likelihoods were reconstructed with the archived dense solver and exact prediction hashes. There are 24 dataset components. Every background-plus-signal vector closes to the original fitted expectation; all expected Poisson counts stay positive. Every grouping matrix contains only zero or one, with no fractional reassignment or reused bin within a panel. The quoted roots and upper endpoints reproduce the frozen dense scan; display grouping never changes a fit.

Mass [MeV]	Dataset	Counts retained in sum	Signal template retained
66	2015	60.1%	87.7%
66	2016	83.6%	96.1%
66	2021	100.0%	100.0%
92	2016	57.8%	83.3%
92	2021	93.8%	98.8%
72	2015	48.0%	77.3%
72	2016	68.0%	88.4%
72	2021	93.1%	98.3%

Table 7: Common-window summed-display retention relative to the complete native fitted window of each year. The full likelihood, amplitude estimates and total-window yield tables use all fit bins. Individual panels have their own resolution-based grouping.

The reusable PDF and PNG figures, native and rebinned arrays, exact grouping maps, exposure tables, build scripts and independent HEP audit are retained in

`study_results/v4p9p16_presentation_extractions_20260906/`

The prior studies were committed, pushed and merged in PR 67; all 4,778 published file blobs were verified on the merged branch. This presentation extension has a separate artifact manifest. The original study inputs and outputs remain frozen.

7 One shared-coupling likelihood

Let $\mathcal{D}(m)$ be the predeclared active dataset set. The model combines Poisson bins and independent Gaussian background constraints:

$$\mathcal{L}_m(\epsilon^2, \theta) = \prod_{d \in \mathcal{D}(m)} \prod_{i \in W_d(m)} \mathrm{Pois}\big(n_{di} \,\big|\, b_{di} + (L_d \theta_d)_i + \epsilon^2 S_{di}(m)\big) \prod_{d \in \mathcal{D}(m)} e^{-\theta_d^{\mathsf{T}} \theta_d / 2}.$$

$S_{di} = \mathrm{BR}_{ee} T_{di}$ is the visible yield per unit physical coupling, where T_{di} is the stored unit-electron-branching template. The fitted coordinate is $\epsilon^2_{\text{eff}} = \epsilon^2_{\text{phys}} \mathrm{BR}_{ee}$; the display restores $\epsilon^2_{\text{phys}} = \epsilon^2_{\text{eff}} / \mathrm{BR}_{ee}$. The implementation rescales this coordinate into a total window amplitude. No independent signal strength is fitted in each year.

An auxiliary unrestricted amplitude supplies the signed likelihood-ratio root,

$$r_m = \mathrm{sign}(\widehat{A}_m) \sqrt{2 \left[\mathrm{NLL}_m(A = 0) - \mathrm{NLL}_m(\widehat{A}_m) \right]}.$$

Here $\mathrm{NLL}_m(A)$ is the negative log likelihood minimized over the background nuisances at fixed A. Poisson expectations remain positive. Physical discovery and upper-limit tests retain the nonnegative signal boundary; the auxiliary negative branch is needed to model the signed field.

Complete joint experiments

Ten pilot and 1,000 separate validation spectra per year are reused from v4.9.14/v4.9.15. The year-specific random streams are distinct. Pairing equal row indices gives 1,000 joint experiments, not 3,000. Each experiment supplies exactly the same full spectrum at every mass in which its dataset participates.

Kernels, templates, support and exclusion masks remain frozen. Every toy retrains its log-count background prediction and count-dependent training errors. All multi-dataset intervals receive new common-amplitude fits. Only the single-dataset intervals reuse existing likelihood-root columns.

The generating means are the archived smooth stress spectra. Event-level independence of the 2016 development input from its full observed sample has not been established. Independence here describes the conditional toy model; omitted experimental systematic correlations have not been newly quantified.

One response basis across boundaries

Following Ananiev and Read [1], perturb each data bin by its Poisson standard deviation. This extension retains the nonzero offset and response width:

$$a_m = r_m(B), \quad D_{im} = r_m(B + \sqrt{B_i} e_i) - a_m, \quad C = D^{\mathsf{T}} D, \quad s_m = \sqrt{C_{mm}}, \quad K_{mn} = \frac{C_{mn}}{s_m s_n}.$$

The basis has 484, 720 and 422 bins from 2015, 2016 and 2021. An inactive dataset has exactly zero response at that mass. This retains correlations across boundaries that share data; separately sampled segment maxima are not joined.

For each statistic, 200,000 vectors $Z \sim N(0, K)$ generate $r_m^* = a_m + s_m Z_m$. The principal maximum is $T = \max_{m: r_m^* > 0} Z_m$ (or $-\infty$ for an empty set); the secondary maximum is $R = \max_m \max(0, r_m^*)$. Their respective tails are compared with the independent joint Poisson scans.

8 Offsets and widths of the combined statistic

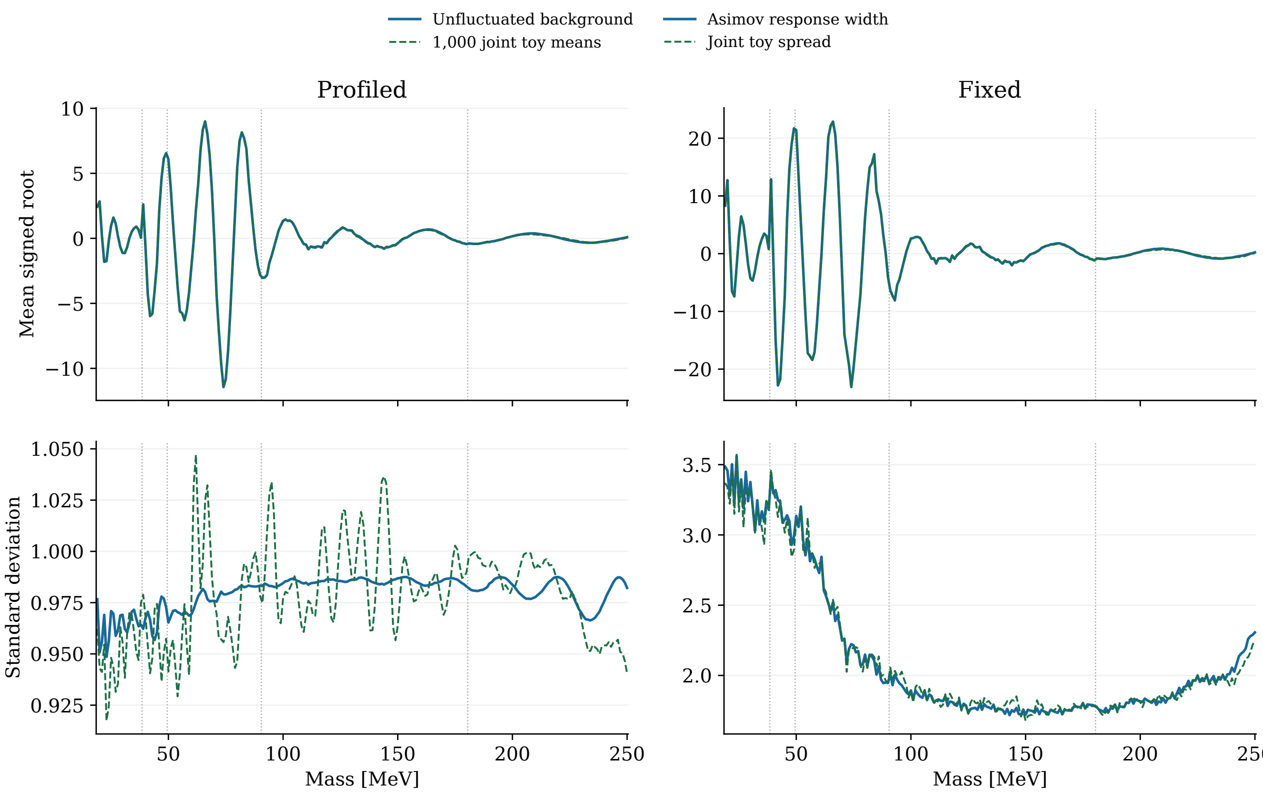

Figure 16: Asimov-response predictions compared with 1,000 joint Poisson scans. Top panels show the offset in the signed root; bottom panels show its spread. Vertical lines mark membership changes. Agreement tests the approximation under the generating means, without establishing their physical validity.

Statistic	Standardized mean range	Spread range	Flags
Profiled	-0.079 to +0.076	0.957 to 1.074	0
Fixed	-0.093 to +0.080	0.943 to 1.050	0

Table 8: Normality flags use a Holm adjustment across 232 masses within each method. Non-rejection has finite power and cannot certify extreme Gaussian tails.

Centering, scaling and covariance come from deterministic bin perturbations. Validation scans do not tune them. Profiled and fixed-background statistics use identical spectra but can have different sampling widths.

Large oscillating offsets can make an ordinary observed value unusual relative to a particular stress construction. A small probability can diagnose that construction without identifying a particle. The inherited 2016 numerical exception, source-fit waiver, development overlap and 75–85 MeV source transition remain unresolved by this extension.

9 Correlations and the look-elsewhere check

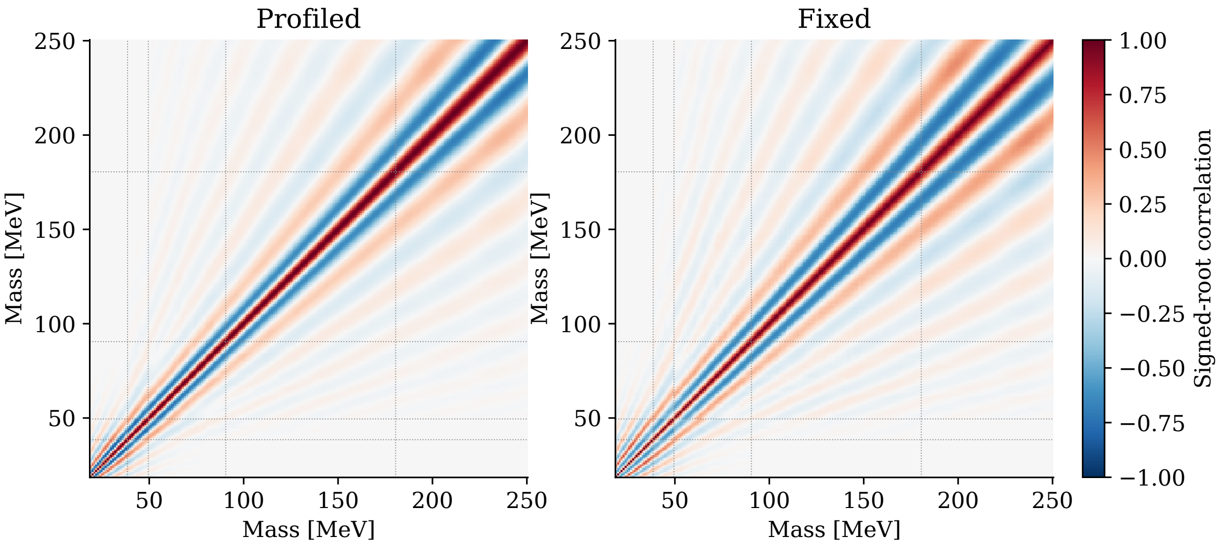

Figure 17: Response correlations across the union grid. Dotted lines show membership changes. Shared datasets retain correlations across boundaries; regions with no common dataset have zero response covariance in this model.

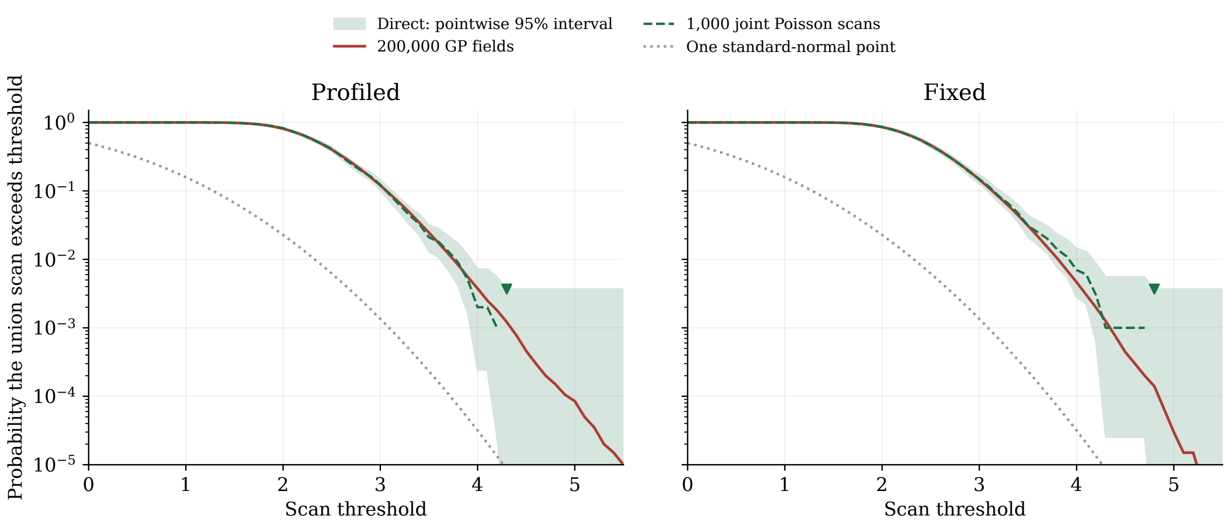

Figure 18: Principal scan-maximum tails. Direct-toy bands are pointwise central 95% binomial intervals, not expected-limit bands or simultaneous coverage bands. Triangles mark the upper endpoint of the central interval at the first unresolved direct tail. More GP draws reduce sampling error within the approximation, without replacing direct validation.

Profiled: the maximum-distribution KS diagnostic gives nominal p=0.8625 for the principal ordering and 0.6389 for the raw ordering. The correlation RMS difference is 0.031.

Fixed: the maximum-distribution KS diagnostic gives nominal p=0.9974 for the principal ordering and 0.2034 for the raw ordering. The correlation RMS difference is 0.030.

These diagnostics compare simulation methods, not the data significance. Two methods are checked per ordering. Direct Poisson counts calibrate the declared score under the generating spectrum without requiring a jointly Gaussian field; the GP estimate has that additional approximation.

10 Representative conditional probabilities

For positive raw roots, the principal local rule is $p_m = \overline{\Phi}[(r_m - a_m)/s_m]$; otherwise $p_m = 1$. Its global curve counts how often any of the 232 masses reaches that local threshold. The separate raw ordering scans $\max_m \max(0, r_m)$. These tests cannot be selected according to which gives the smaller probability.

Mass	Active years	Nominal local	Ref. Gaussian	Ref. GP global	Direct	Raw global
30	2015	0.1291	0.01	0.5479	543/1000	1
65	2015+2016+2021	0.0052	1	1	1000/1000	1
66	2015+2016+2021	0.0029	1	1	1000/1000	1
76	2015+2016+2021	0.4342	6.98×10^{-20}	$<1.50 \times 10^{-5}$	0/1000	1
120	2016+2021	0.0795	0.0598	0.9813	975/1000	1
220	2021	0.5	1	1	1000/1000	1

Table 9: Profiled representative masses: the principal and raw extrema, plus 30, 65, 120 and 220 MeV. Ref. Gaussian is a formal local tail of the reference-root approximation, not a validated particle probability. Ref. GP global adds the scan maximum under the same reference. Both global columns use the full union at the threshold observed at that mass. Raw global is the separate raw-peak test.

Statistic	Mass	Raw r	Offset a	Width s	$(r-a)/s$
Profiled	76	+0.166	-8.700	0.979	+9.053
Fixed	76	+0.175	-15.348	2.164	+7.173

Table 10: Largest reference-centered score decomposition. The final column is a conditional score, not an independently established discovery significance.

At the largest profiled reference-centered score, the unadjusted asymptotic local probability is 0.4342, while centering and scaling against the stress background gives 6.98×10^{-20}. The observed raw root is close to zero; the extreme centered score arises mainly because the stress background predicts a large negative root.

At 74 and 75 MeV, the centered scores are 10.20 and 10.38, but the raw roots are negative. The declared boundary therefore assigns local p=1 there. The next positive raw root becomes the reference-centered extremum. This abrupt change comes from the signal boundary together with the stress offset; it is not a numerical sign flip.

The raw maximum occurs at 66 MeV. All 200,000 GP fields and all 1,000 direct experiments exceed it. These differences reflect the specified orderings and background offsets; they are not a free choice of significance. The raw maximum tests positive peaks; it does not assess full-spectrum goodness of fit.

Statistic	Full 1 MeV grid	First 2 MeV subgrid	Second subgrid
Profiled	$<1.50 \times 10^{-5}$	$<1.50 \times 10^{-5}$	$<1.50 \times 10^{-5}$
Fixed	$<1.50 \times 10^{-5}$	$<1.50 \times 10^{-5}$	$<1.50 \times 10^{-5}$

Table 11: GP probabilities at the same original peak threshold. Zero-count bounds do not establish fine-tail convergence.

Subgrids remove alternate masses at the original observed threshold. They test finite-grid dependence, not convergence to a continuous search. Probabilities include neither a choice between methods nor uncounted searches over widths, kernels or other analysis choices.

11 Observed limits and the v4.9.12 comparison

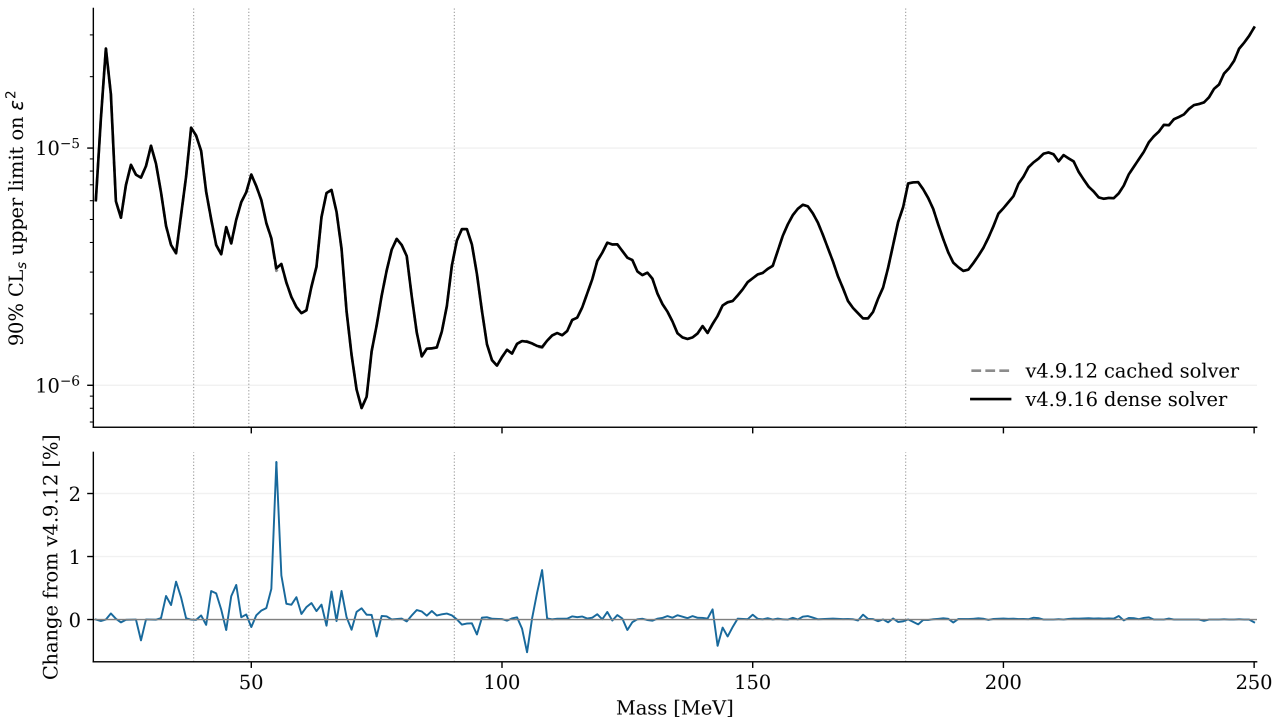

Figure 19: The pointwise upper-limit construction with the current dense and earlier cached solvers. The lower panel shows their relative difference. Both curves have the same membership and branching correction. Their difference is not a look-elsewhere adjustment or a sensitivity measurement.

The largest absolute relative change from the corresponding v4.9.12 limit is 2.502%. The largest difference in the nonnegative discovery root is 0.0267. Fresh dense fits reproduce v4.9.13 dense references where the same scope exists. The old optimizer does not select a preferred answer. Above 211.317 MeV, displayed limits include $1 + \sqrt{1 - 4m_\mu^2/m^2}(1 + 2m_\mu^2/m^2)$ once; the CSV retains the electron-channel values.

The upper limit inverts the bounded, piecewise-asymptotic CL$_s$ statistic at each mass [2]. The background-only joint scans test the discovery-score approximation. They do not supply the signal-plus-background distributions needed to calibrate the full CL$_s$ curve. A GP global probability is not a replacement CL$_s$ tail.

The v4.9.13 study contains conditional calibrated individual limits and the common all-three interval. It has no calibrated pairwise endpoints over the remaining union. Splicing different calibrations would conceal a change of construction. A fully calibrated union limit requires additional declared signal-plus-background work; this note supplies the complete observed asymptotic curve.

12 Numerical checks, provenance and continuation

All 2736 numerical and product-identity checks passed. There are 142 newly fitted joint coordinates and 90 reused single-dataset coordinates. Summed coordinate execution took 24.1 minutes with one worker and one BLAS thread. The exact backend was retained at 1 new coordinates (41 MeV). The largest paired pilot-root difference is 2.10×10^{-5}, the largest complete-response relative vector error is 4.73×10^{-6}, and the largest sentinel-correlation error is 4.65×10^{-7}. No paired pilot changed its raw-positive classification.

Every new joint mass has ten paired exact pilot fits, an exact baseline and a fixed bin-response stencil. Complete exact responses are retained at 39, 49, 50, 90, 91 and 180 MeV. The accelerator must pass root, centered-response, width and correlation gates. Otherwise all ensembles at that coordinate use the exact backend. Failed exact references halt; no toy or hypothesis is removed.

The v4.9.14 and v4.9.15 inputs are checked against their full frozen manifests. The v4.9.12 comparison table is checked against its specific frozen-manifest entry. The isolated derivative is

`study_results/v4p9p16_combined_global_20260906/`

It retains spectra, joint root vectors, limits, covariance, GP maxima, probability tables and per-mass audits. The independent HEP review is under `review/`. The original manifest remains unchanged. The frozen deficit extension has a separate manifest under

`study_results/v4p9p16_deficit_extension_20260906/`

The independent deficit audit passed 939 conditions.

From the repository root, verify the completed numerical products with

```
python3 -B \
  study_results/v4p9p16_combined_global_20260906/verify_combined.py
```

Preserve the finished version before reanalysis or new generation. `NEXT_STEPS.md` gives resume commands, new-seed requirements and the additional work for calibrated limits.

The next scientific priority is to qualify the joint background family with predeclared predictive controls. Direct tail precision, finer grids and signal injections answer separate questions. Additional GP sampling alone cannot establish physical background validity, expected sensitivity or interval coverage.

The probability audit and current-solver echo replay are retained separately in

`study_results/v4p9p16_probability_echo_review_20260906/`

They preserve every original scan value and source manifest. The new probability ledger contains all 232 masses, exact exceedance counts, and pointwise Monte Carlo intervals. The deterministic echo archive stores fitted expectations and nuisance components for independent reconstruction. No new events were unblinded and no new random toys were generated for this revision.

13 Exploratory 2015 search on the 15–20 MeV rising edge

Result. This short-support study does not reveal a persuasive narrow excess in 15–20 MeV. The largest nominal-GP upward fluctuation is at 17.25 MeV. Releasing an active smoothness ceiling gives a stable fit with $r = 1.58$ and local asymptotic $p_0 = 0.057$; the independently fitted polynomial background gives $p_0 = 0.254$ there. These values are model-conditional, selected from a scan, and do not establish or exclude a heavy photon.

Why a separate support is useful

The released 2015 scan begins at 19 MeV, with GP training support 14–135 MeV. Its input histogram extends below that support. The spectrum contains 81 events in 12–13 MeV, 7,653 in 14–15 MeV and 330,090 in 19–20 MeV. A local GP can isolate this turn-on without fitting the distant falling spectrum. It must still describe the steep background accurately across a missing signal window.

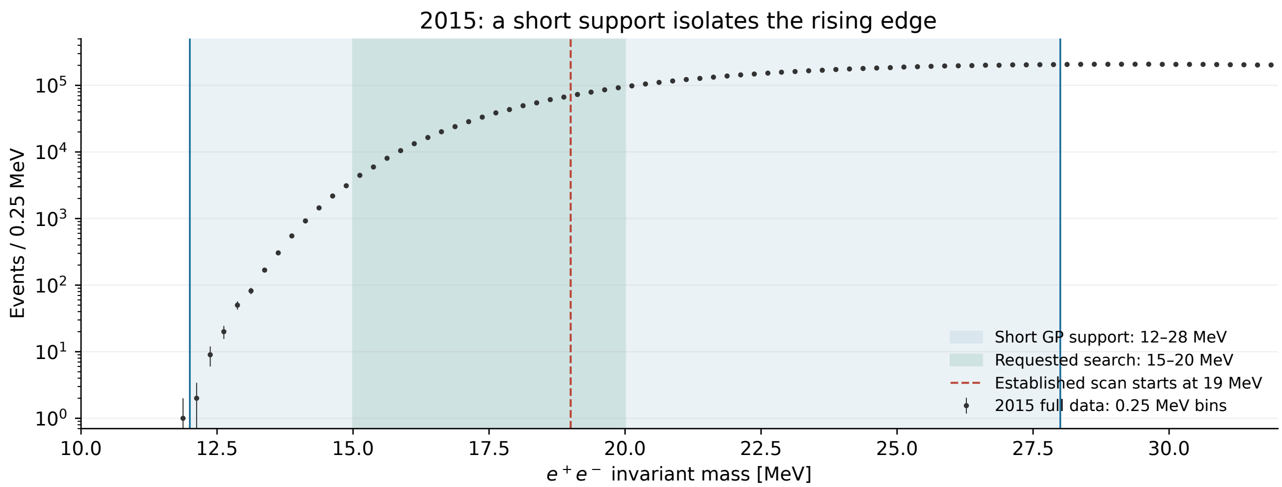

Figure 20: The same released full-2015 histogram, summed from 0.05 to 0.25 MeV bins. The chosen GP support is 12–28 MeV; the requested search is 15–20 MeV. The dashed line marks the lower boundary of the established scan, not the lower boundary of the available counts.

The new scan has 21 hypotheses, separated by 0.25 MeV. A labelled continuation to 22 MeV connects it to the previously selected 21 MeV point; that continuation is excluded from selection of the leading 15–20 MeV feature. The comparison supports 12–26, 12–30 and 12.5–28 MeV were declared before examining the new signal fits. Their p-values do not select the nominal support.

Detector boundary. The inherited Gaussian width is $\sigma_m = -0.09223 + 0.053219m$ MeV for m in MeV, giving about 0.71–0.97 MeV here. Below 19 MeV it is an extrapolated shape assumption. The published 2015 prompt search covered 19–81 MeV [3]. Low-mass signal efficiency, acceptance and the selected signal shape need direct simulation and control checks before a physical coupling result can be quoted.

Local probabilities and dependence on the background

The GP fits log counts versus log mass outside $m \pm 2.25\sigma_m$. Its prediction and correlated count covariance constrain the background in an exact Poisson window likelihood. Both the signal amplitude and background nuisance coordinates are fitted; the auxiliary amplitude may be negative while every total expectation remains positive. We report

$$r = \operatorname{sign}(\widehat{A})\sqrt{2[\mathrm{NLL}(0) - \mathrm{NLL}(\widehat{A})]}, \qquad p_0 = \overline{\Phi}(\max(r, 0)) .$$

Thus the asymptotic convention assigns $p_0 = 0.5$ to negative fits [2]. No GP estimate of a global trials factor is used in this section.

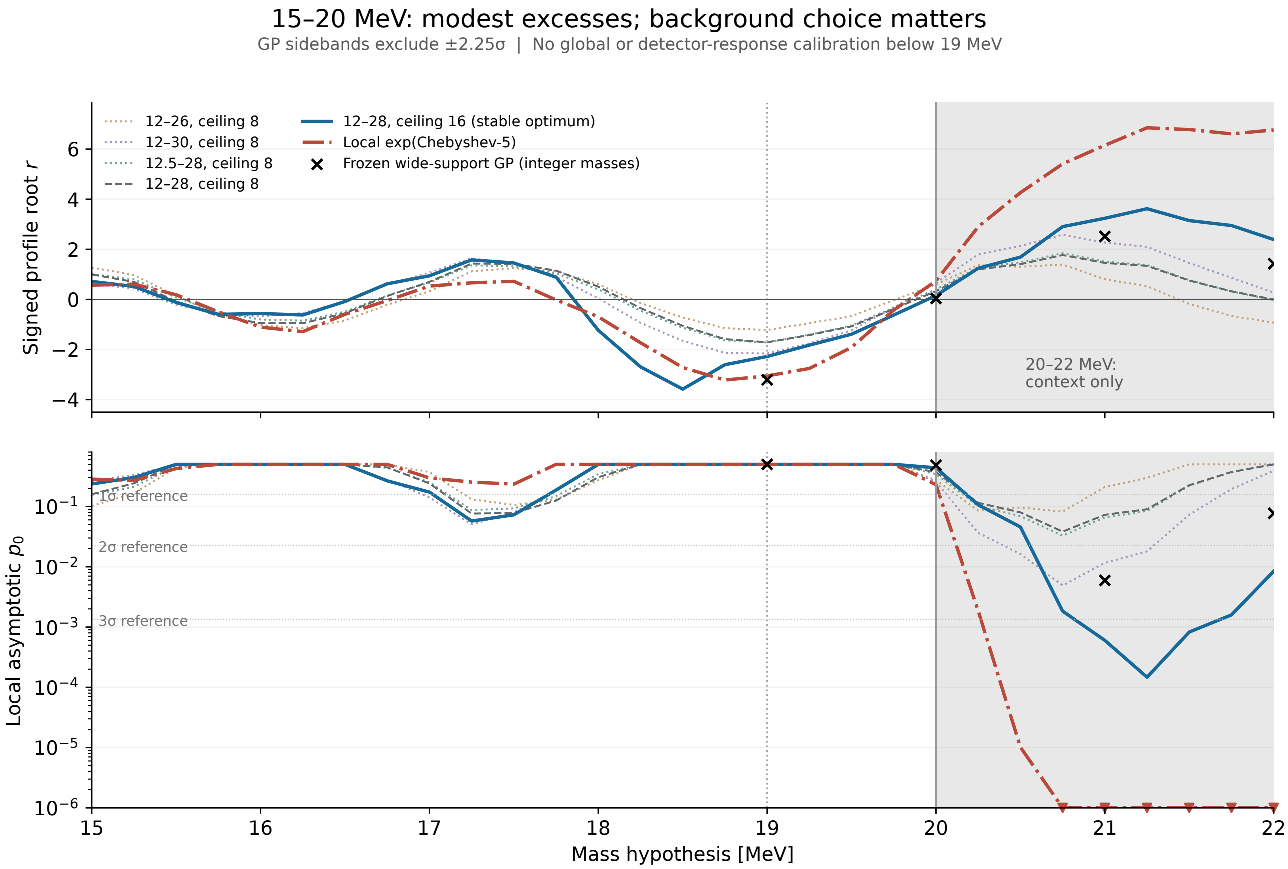

Figure 21: Signed roots and local asymptotic excess probabilities. All four initial GP supports reach the inherited ceiling of eight resolution units. Raising the nominal-support ceiling to 16 removes every active boundary; ceilings 32 and 64 recover the same optima, with maximum root changes below 0.00054. The local polynomial is a separate background-family check. Gray shading identifies the 20–22 MeV bridge. Triangles mark the 10^{-6} display floor, not resolved global tails.

For the polynomial check, a positive exp(Chebyshev$_5$) background and Gaussian signal are integrated over each 0.25 MeV bin. All background coefficients are profiled over a moving $\pm 7\sigma_m$ support, clipped at 12 MeV. Its form and window scale follow the published low-mass procedure [3]; its suitability below 19 MeV is tested here rather than assumed. In 15–20 MeV its largest positive root is only 0.74. At 21 MeV it gives a much larger root, 6.15, but a signal-plus-background Poisson deviance of 98.7 for 51 nominal degrees of freedom. That poor-fit diagnostic and the GP dependence prevent interpreting the large value as a robust signal.

Signal extraction at fixed low-mass anchors

The displays use the stable ceiling-16 GP on 12–28 MeV. The 15 and 17 MeV hypotheses were fixed before the scan. Curves for profiled components are confined to the fitted window. Outside it, only data and the sideband GP constraint are shown. The gold uncertainty is the GP constraint before fitting the window, not an uncertainty band for the final fitted background.

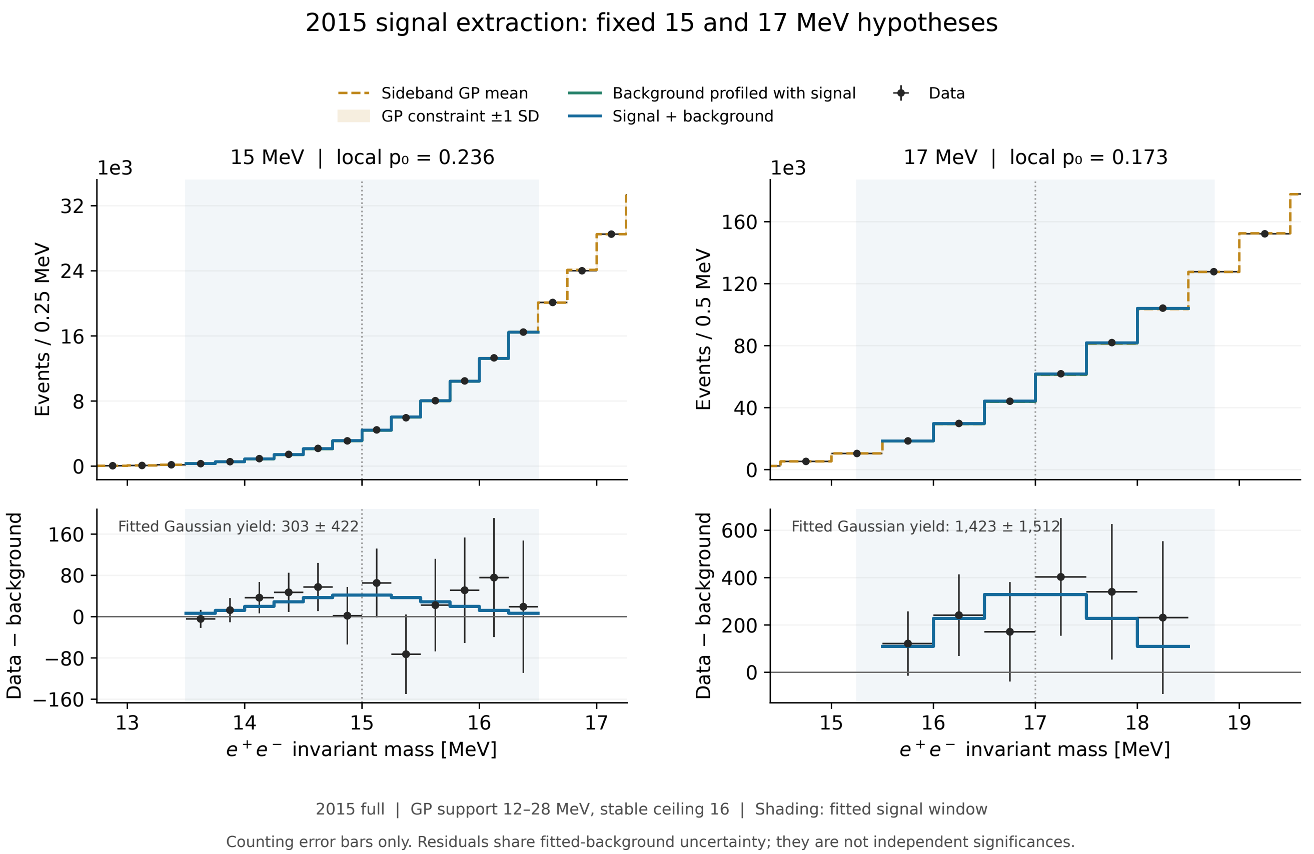

Figure 22: Fixed 15 and 17 MeV hypotheses. Top: data, sideband background constraint, profiled background and signal-plus-background prediction. Bottom: data minus the background profiled with signal, with the fitted Gaussian contribution. The error bars describe counting uncertainty only. The subtraction uses the same data to fit the background and is not an independent pull test.

The likelihood uses 0.25 MeV bins. For these figures, complete groups of bins are chosen nearest to half a mass resolution and anchored at the original zero edge. The resulting display widths are 0.25 MeV at 15 MeV and 0.50 MeV at the other displayed masses. Boundary groups crossing the fitted window are omitted from the profiled display, while all original window bins remain in the likelihood. No grouping or phase is chosen to make a peak sharper.

m [MeV]	r, ceiling 8	r, ceiling 16	p_0, ceiling 16	p_0, polynomial	Fitted yield, ceiling 16
15	1.00	0.72	0.236	0.284	303 ± 422
17	0.70	0.94	0.173	0.298	$1,423 \pm 1,512$
17.25	1.43	1.58	0.057	0.254	$2,542 \pm 1,612$
20	0.33	0.16	0.438	0.230	$514 \pm 3,312$

Yield errors are local curvature errors for the assumed Gaussian template; they are not post-selection confidence intervals. No conversion to ϵ^2 is made.

The leading fluctuation and the upper edge of the requested interval

At 17.25 MeV the stable GP fit gives approximately 2,542±1,612 Gaussian-template events. Its maximum is selected after scanning 15–20 MeV, so this amplitude can be upward biased. It does not acquire a stronger interpretation because the scan was restricted to a rising edge. The 20 MeV panel supplies the fixed upper anchor.

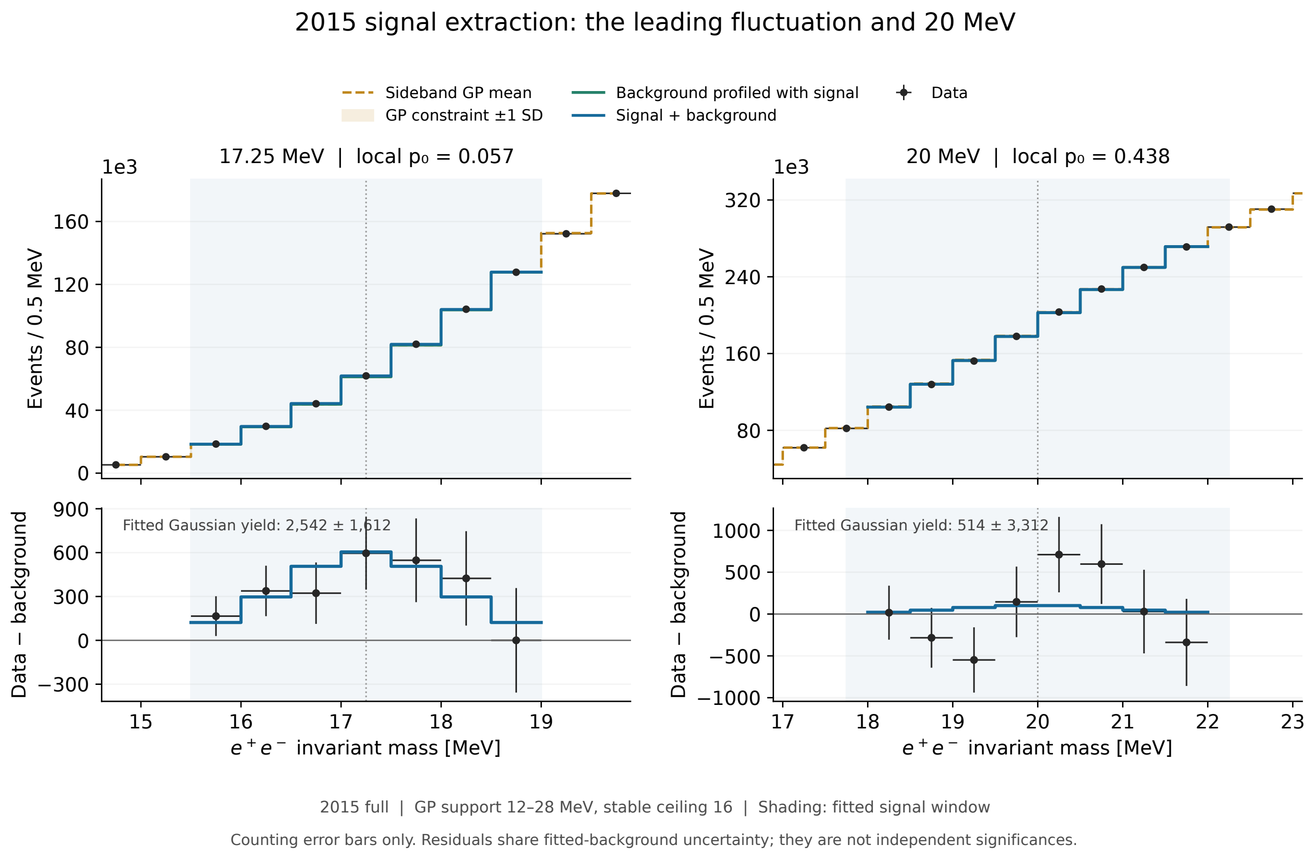

Figure 23: The selected 17.25 MeV upward fluctuation and the fixed 20 MeV hypothesis, with the same conventions and fixed-phase grouping as the preceding displays. Their local asymptotic roots are 1.58 and 0.16. A convincing resonance would need a stable excess across defensible background choices and a validated accepted signal shape.

At 21 MeV, the nominal GP changes from $r = 1.46$ to $r = 3.24$ when its active smoothness restriction is removed; the frozen wide-support fit has $r \simeq 2.52$. This is evidence that the inference is sensitive to the background description. It is not evidence that the low-mass study has uncovered a second established particle feature. Resolving this dependence requires predictive background controls and detector-level checks, not choosing the smallest p-value.

Small conditional toy checks and what remains to establish

Ten pilot spectra were first generated at each of the four displayed masses, then extended to 100 at each mass for each of ceilings 8 and 16: 800 distinct mass/model toy fits in total. The first ten are included in each 100. Each Poisson spectrum covers the complete 12–28 MeV support, and its GP hyperparameters and background are fitted again. The generating mean is the continuous sideband-conditioned GP prediction at that mass, without an observed-window adjustment. These mass-local backgrounds cannot be joined into one global experiment.

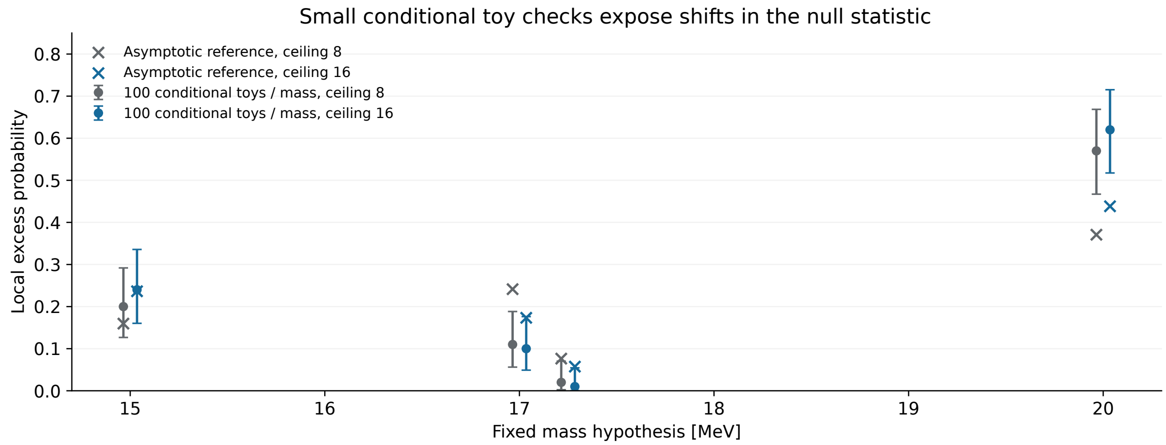

Figure 24: Toy upper-tail fractions with central 95% binomial intervals, compared with the asymptotic local reference. The two ceiling choices have their own conditional generating backgrounds. These small plug-in ensembles assess this fitting procedure under those means; they are not independent background validation or a final p-value calibration.

Ceiling	m [MeV]	Tails/toys	Fraction	95% interval	Mean r	SD r
8	15	20/100	0.20	[0.127, 0.292]	+0.23	0.89
8	17	11/100	0.11	[0.056, 0.188]	-0.48	0.95
8	17.25	2/100	0.02	[0.002, 0.070]	-0.43	0.99
8	20	57/100	0.57	[0.467, 0.669]	+0.52	0.92
16	15	24/100	0.24	[0.160, 0.336]	+0.07	1.04
16	17	10/100	0.10	[0.049, 0.176]	-0.31	0.97
16	17.25	1/100	0.01	[0.00025, 0.054]	-0.38	0.94
16	20	62/100	0.62	[0.517, 0.715]	+0.42	1.01

The shifts of the toy-root means away from zero demonstrate why an asymptotic reference alone is insufficient to qualify this extension. A small toy-tail fraction is conditional on a data-derived mean and has finite sampling uncertainty. It does not remove the choice of mass, support or background family.

Next, validate accepted signal templates and the turn-on; test background prediction and signal recovery under alternative smooth truths; fix choices independently of signal-window residuals; and calibrate a coherent scan for a global probability. A shorter GP support does not remove look-elsewhere accounting. The accompanying study bundle retains scripts, fit components, display maps, 800 toy roots and source hashes.

References

[1] V. Ananiev and A. L. Read, *Gaussian Process-based calculation of look-elsewhere trials factor*, JINST **18** (2023) P05041. arXiv:2206.12328v3.

[2] G. Cowan, K. Cranmer, E. Gross and O. Vitells, *Asymptotic formulae for likelihood-based tests of new physics*, Eur. Phys. J. C **71** (2011) 1554. arXiv:1007.1727.

[3] P. H. Adrian et al. (HPS), *Search for a Dark Photon in Electro-Produced* e^+e^- *Pairs with the Heavy Photon Search Experiment at JLab*, Phys. Rev. D **98** (2018) 091101. arXiv:1807.11530.